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Research Project

Comparison and Optimization of Feature Ranking Methods in Medical Risk Assessment

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Abstract

Feature importance ranking has emerged as a powerful tool for explainable artificial intelligence (XAI), offering insights into model behavior and identification of key predictive features. In the healthcare domain, it aids in identifying the critical risk factors that support early disease detection and preventive care. This study evaluates and compares various feature ranking methods in the context of identifying diabetes risk predictors across two distinct datasets. Common risk factors such as body mass index (BMI), high cholesterol, high blood pressure and age were consistently ranked among the top features across both datasets. Additionally, magnetic resonance imaging (MRI) based fat fraction measurements in gluteus thigh muscles were also identified as an important biomarker for diabetes onset.

The quality of feature rankings was assessed by training downstream classifiers on varying subsets of top ranked features. Feature mask (FM) module based ranking demonstrated superior performance across various classifiers. Notably, most ranking methods achieved peak accuracy values using only the top 30% of features, beyond which performance plateaued and in some cases slightly declined. This validates the informativeness of the top ranked features.

Furthermore, pairwise Spearman correlation analysis of feature rankings revealed moderate agreement (values greater than 0.5) among most methods. This insight is crucial for biomarker discovery, as it encourages the integration of different ranking techniques for more comprehensive feature selection. The findings contribute to model interpretability and to enhance trust in AI-driven disease risk prediction, supporting its adoption in clinical settings.

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1. Introduction

Machine learning (ML) and deep learning (DL) models have become foundational tools for detecting patterns and making predictions from complex data. Among various forms of data encountered, tabular datasets are the most common. These datasets are structured collections of features and target variables organized in rows and columns. They serve as the backbone for many critical applications in finance, logistics and healthcare domains. However real-world tabular datasets presents their own set of unique challenges. High-dimensional feature spaces, missing or noisy values, complex inter-feature dependencies and heterogeneous data types often complicate the learning process [1]. Moreover, such real-world datasets include numerous irrelevant or noisy features that can obscure useful patterns and lead to poor model generalization. In such cases, where sample size is dwarfed by the number of measured features, models trained on such data may suffer from high variance, increased training times, and reduced interpretability.

When making a choice for a university to attend, students typically compare available programs, tuition fees, academic reputation, campus facilities, industry interactions and social life. Naturally, different students will prioritize different aspects. One student may focus more on academic rankings, available programs and ongoing research areas. An international student might prioritize more on campus facilities, support services to international student. On the other hand, a student with financial constraints may focus mainly on tuition fees and available scholarships. From the above example we can see that not all features are given equal importance and they vary based on different objectives.

While data practitioners often curate important features for a particular problem based on domain knowledge, this ability is limited when presented with hundreds to thousands of features. In such scenarios, we make use of the feature ranking methods available to select important features for model predictions. Feature selection plays an important role in modern machine learning pipelines. Feature selection is the process of identifying and retaining the most informative features while eliminating those that contribute little to the predictive task. It helps reduce overfitting, improve model performance, and enhance interpretability [2]. It is particularly important in high-stakes decision-making contexts like healthcare where effective feature selection can improve clinical decision support, diagnostic accuracy and patient outcomes.

1.1. Motivation

With increasing availability of data in the health care industry, Artificial Intelligence (AI) is used for various purposes such as disease diagnosis, treatment, imaging analysis and drug development. AI provides medical practitioners with important insights that help in improving health outcomes and patient experiences.

Risk Assessment is one important application that predicts the likelihood of a patient developing a disease or condition in the future based on various risk factors such as genetics, lifestyle, environmental exposures, and existing health conditions [3]. This helps in identifying populations that are at high risk for a certain disease and provide them with targeted preventive measures like lifestyle modifications, regular screenings and early medical treatments. This reduces the burden of a disease at both individual and population levels. However, effective risk assessment depends on identifying the right set of features such as clinical biomarkers, behavioral attributes and lifestyle patterns. Without knowing which variables contribute meaningfully to a condition's onset, predictive models risk being inaccurate, overly complex, or uninterpretable to healthcare providers. This highlights the importance of feature ranking, which plays a crucial role in enhancing the model interpretability and ensuring clinical trustworthiness.

Diabetes mellitus was selected as the target condition for this study due to its growing global prevalence and significant public health implications. The International Diabetes Federation estimates that by 2045, at the current growth rate, 693 million people will have diabetes worldwide. The disease is particularly concerning because its onset often goes undetected until advanced stages leading to severe complications such as cardiovascular disease, kidney failure, or vision loss [4]. From a machine learning perspective, diabetes risk prediction offers a compelling case study owing to the availability of high-quality tabular datasets and the wealth of ongoing medical research which provides a strong basis for comparing and validating computational findings against established domain knowledge.

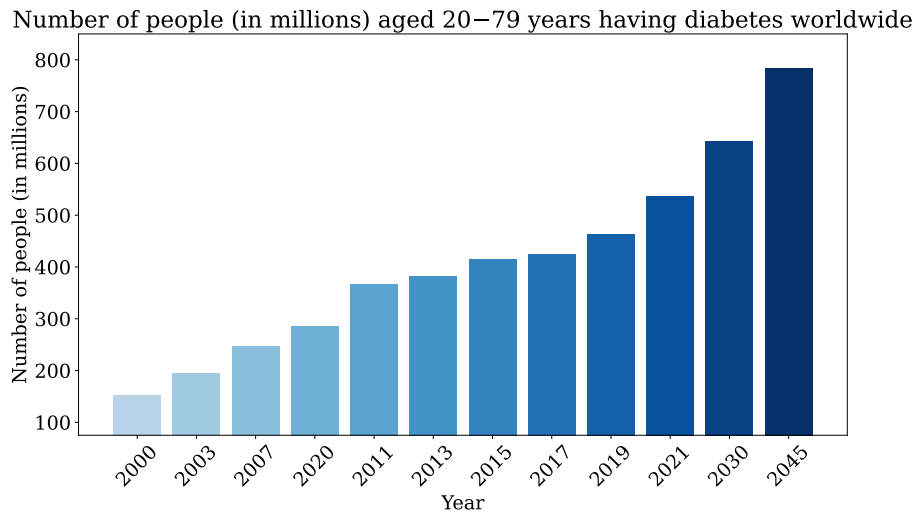


Figure 1.1.: Global trend in the number of people aged 20–79 years with diabetes (in millions) from 2000 to 2045. The data shows a significant and steady increase (Year 2030 and 2045 have projected numbers) [4]

There are 3 types of diabetes: type 1, type 2 and gestational. The most common type of diabetes among these three is the type 2 diabetes (diabetes mellitus) which accounts for almost 90% to 95% of all cases. It is important to note that type 2 diabetes is a lifestyle disease which usually occurs mainly due to poor diet, low physical activity, obesity and

age. This makes diabetes a predictable and preventable disease by detecting the risk factors associated with it. [4]

1.2. Research Goals

Motivated by the necessity for early detection of diseases in the healthcare industry this study aims to identify the potential behavioral, lifestyle and biological risk factors for diabetes prediction. Further, to enable reliable risk assessment and to understand the predictive models better, different feature ranking techniques are compared and its impact on downstream classification models are evaluated.

1. What are the most important features that can be considered as medical risk factors for the onset of diabetes?
2. How do the top ranked features impact the performance of the downstream model?
3. How similar are the different feature ranking results?

To answer the above questions, we establish a feature ranking and evaluation framework that integrates multiple feature ranking methods with downstream classification models. This enables a systematic comparison of different methods and analyze their agreement of identifying common influential features consistently across multiple methods. We evaluate how different subsets of selected features influence model performance by assessing classification accuracy. Specifically, we investigate how the number of selected features impacts the performance of the downstream model and whether adding more features leads to potential overfitting. We also explore which feature ranking methods are most effective when paired with specific classifiers, providing insights into the compatibility between selection strategies and learning algorithms.

2. Feature Ranking and Selection Methods

Features are the measurable quantities or attributes that characterize the data. In medical datasets, features such as age, gender and blood pressure provide information about the health status of individual patients. These features may be categorical, discrete numerical and continuous numerical values.

A discriminative classifier is a model that learns the posterior, which is the probability of target class y , given the input variable x from the data and predicts the target class of new instances [5]. They learn to distinguish between the classes without modeling the underlying distributions of the features. Consider an example of classifying the fruits based on its observable characteristics like color and shape. The model might learn that if a fruit is yellow and long, the probability that it is a banana is extremely high. Conversely, if it is round and red, it is most likely an apple. This decision-making process is guided by the model's ability to draw decision boundaries based on the features. If color better separates the classes, then the model will rely more on that feature when making new predictions.

Feature ranking methods quantify the contribution of each feature to the model's discriminative power by assigning importance scores. These importance scores can be used to generate an ordered list of features, which can then be used to select an optimal subset of features for building effective predictive models. Overall, feature ranking and feature selection are closely related processes in identifying the most relevant features for predictive modeling. This work focuses on how different ranking strategies influence the selection of relevant features and the resulting predictive performance of classifiers trained on these selected features.

Formally, let each data point be represented by a d -dimensional feature vector, $X = [x_1, x_2, \dots, x_d]^T \in R^d$, where x_i denotes the value of the i -th feature. Feature ranking and selection aims to select a subset of important features $F \subset X$ that effectively represent the original data while maintaining a strong predictive performance on the classification task. The goal is to retain features in F that meaningfully contribute to predicting the target variable y i.e.,

$$S = X \setminus F = [x_1, \dots, x_i, \dots, x_p]^T$$

A feature x_i is considered relevant if its presence changes the conditional distribution of the target variable, if and only if there exists x_i, y and s_i for which $P(x_i = x_i, s_i = s_i) > 0$ such that [6],

$$P(y = y_i | s_i = s_i, x_i = x_i) \neq P(y = y | s_i = s_i) \quad (2.1)$$

Conversely, a feature x_i is considered irrelevant if it does not influence the distribution of y given S . That is,

$$P(y = y_i | s_i = s_i, x_i = x_i) = P(y = y | s_i = s_i) \quad (2.2)$$

For instance, in the previously discussed fruit classifier, in addition to features such as color and shape, a unique Fruit ID is assigned to each instance. In this case, color and shape are relevant features that contribute in distinguishing between different fruits. In contrast, Fruit ID serves only as a unique identifier and does not contain any information useful for classification, making it an irrelevant feature.

Most real world data are high dimensional and often contain noisy and irrelevant features that can hinder the model performance. Feature selection based on feature importance is used to address the curse of dimensionality [7] and improve generalization of the models. Beyond aiding in improving model performance, feature importance ranking also contributes to a better understanding of model behavior and aids in discovering key features for a task.

2.1. Related Work

Numerous studies have investigated disease risk prediction, particularly for chronic conditions such as diabetes mellitus by analyzing risk factors and identifying significant predictors using machine learning models. Publicly available datasets like the Pima Indian Diabetes Dataset and the Behavioral Risk Factor Surveillance System (BRFSS) have been extensively used in this context. These studies [8, 9] extract medically relevant insights by identifying influential features contributing to the disease onset. These studies incorporate statistical measures such as odds ratio and dimensionality reduction techniques like Principal Component Analysis (PCA) to identify major risk factors.

Feature selection techniques emerged as the widely adopted strategy for identifying the most influential features and to address the curse of dimensionality. Most feature selection methods aim to identify feature subsets through pre-defined criteria to optimize the performance of learning systems. Feature importance ranking (FIR) is often one of the first step used when selecting a subset of features for modeling. Feature ranking is a subdomain of feature selection which aims to assign an importance score to each feature to facilitate selection or interpretation [10]. Ranking-based methods offer finer control and interpretability compared to simple binary inclusion or exclusion of a feature as used in selection.

Studies on feature importance ranking like [11, 12] leverages XGBoost algorithm to rank features for identifying risk factors relating to heart conditions. Deep learning based ranking methods on medical datasets have been explored in [13]. Some works, such as [14, 15] focused more on the interpretability where importance rankings were compared with explainability techniques like SHapley Additive exPlanations (SHAP) analysis or such techniques were employed separately. In addition to ranking-based methods, many studies have explored traditional feature selection techniques, typically categorized into filter, wrapper, and embedded methods. Some of these studies in the context of medical risk identification are discussed below.

- Filter methods [16, 17, 18] select features independently of any learning algorithm and rely on statistical properties of the data. Features are ranked based on their relationship to the target variable, quantified based on correlation coefficients, Fisher discriminant ratio, scores from t-tests, chi-squared tests and measures from information theory such as entropy or mutual information. While filter methods are computationally efficient,

they tend to ignore the dependencies and interactions among the features. Recent improvements attempt to consider pairwise interactions [19], but they may still select redundant or suboptimal feature subsets, particularly in complex domains such as genomics or multi-modal sensor data.

- Wrapper methods [20, 21] incorporate the learning algorithm into the feature selection process. These methods search the feature subspace iteratively and the candidate subsets are evaluated on the learning algorithm to identify the optimal combination of features. Naively speaking, if there are m features, then a total of 2^m possible subsets needs to be evaluated. Therefore, iteratively training on subsets of features might be computationally expensive and in order to address this many optimization techniques such as forward selection, recursive feature elimination and genetic algorithm based searches are studied to mitigate the computational cost [22, 23]. However, in these methods there is no guarantee that selected features on a certain learning algorithm will remain optimum on another.
- Embedded methods [24] integrate feature selection directly into model training process optimizing both feature relevance and model parameters simultaneously. The search for optimum feature subset and optimum parameters for the predictive model becomes a joint optimization problem. These include decision-tree based algorithms, regularization based feature selection and more recent approaches such as incorporating attention mechanisms for feature selection.

While many of these studies effectively utilize feature ranking or selection methods to identify the significant variables, they often fall short in evaluating the impact of the selected features on the predictive performance of the model. In other words, these works often stop short of validating whether the identified features lead to better performance in prediction tasks, which is crucial for real-world deployment of AI systems in healthcare. Studies like [25, 26] present a comparative evaluation of feature selection strategies for cancer and heart disease predictions. They provide comparisons between filter, wrapper and embedded based approaches demonstrating an average performance improvement of 2% in model accuracy when applying feature selection. While individual feature selection techniques have been applied to diabetes prediction in the past studies, a systematic comparison of multiple methods, their agreement, and their influence on different machine learning classifiers remains unexplored. Furthermore, the consistency of selected features across methods and the performance implications of varying feature subset sizes have not been comprehensively studied.

This study aims to address these gaps by a unified evaluation framework to compare and assess multiple feature ranking or selection methods on tabular healthcare datasets. In this study, we focus on an information gain based method, a tree based importance scoring method and a couple of deep learning based feature selection methods. By integrating these methods with downstream classifiers, we aim to evaluate their predictive performance and examine the agreement across ranking strategies. This comprehensive analysis contributes to the interpretability, robustness, and practical deployment of machine learning models in healthcare risk assessment.

2.2. XGBoost Feature Importance

XGBoost (Extreme Gradient Boosting) [27] is an ensemble learning technique in which multiple weak learners are trained sequentially to build a strong predictive model. Each weak learner is a shallow decision tree which may result in poor predictions individually. However, when combined sequentially using a boosting framework, these weak learners form a robust and accurate model. In boosting, trees are built sequentially where every new tree is trained to improve upon the errors made by the previous tree.

This process can be intuitively compared to a sports team improving its performance over time by learning from its past mistakes. After each game (iteration of model) the coach analyses the mistakes (errors or residuals) and makes strategic improvements (updates to model parameters). Instead of retraining the entire team from scratch, each session focuses on correcting the mistakes from the previous game. Similarly, each decision tree acts as a training session, and collectively, the model improves iteratively to reduce prediction error (loss function).

The feature importance for a gradient boosted model is computed by averaging the importance calculated across all the trees in the ensemble. Within each tree, feature importance can be calculated based on the features used at the split nodes. This is conceptually similar to how discriminative models operate. There are three prominent measures to calculate this importance:

- Weight- is based on the number of times a feature is used to split the data in a tree.
- Gain- evaluates the improvement in the loss function when a feature is used for a split.
- Cover- enumerates the total number of observations affected by a feature when it is used in a split. This indicates how broadly a feature influences the data split.[28]

While weight and cover show how frequently or broadly a feature is used, they do not necessarily reflect its contribution to improving the model's predictions. In this work, we rely on gain as the metric for calculating feature importance, as it more directly captures a feature's impact on the model's predictive performance.

2.2.1. Gain Based Feature Importance

To understand how feature importance is calculated using gain criterion, it is important to understand how each tree in the ensemble is built. This section outlines the fundamentals of tree construction and how feature importance is derived from the gain criterion.

Each tree in a tree ensemble is a Classification And Regression Tree (CART) which is a binary tree. At each internal node of the tree, data is split into two disjoint subsets based on a threshold value of selected feature. Every instance in the dataset $x_i \in R^m$ is classified based on the split criteria at the nodes until it reaches the leaf node where it is assigned a label y_i and a score $w \in R^T$. This can be represented as, $q : R^m \rightarrow T$ where q is the structure of each tree and T is the number of leaves. Unlike standard decision trees that only produce class labels, CART assigns a continuous score to each leaf, aiding both in prediction and optimization.

Mathematically, we can express the predicted output label as sum of outputs from all K trees in the ensemble:

$$\hat{y}_i = \phi(x_i) = \sum_{k=1}^K f_k(x_i), f_k \in F, \quad (2.3)$$

where each function f_k represents a tree, defined as $f_k(x) = w_{q(x)}$ and F is a space of all possible CARTs.

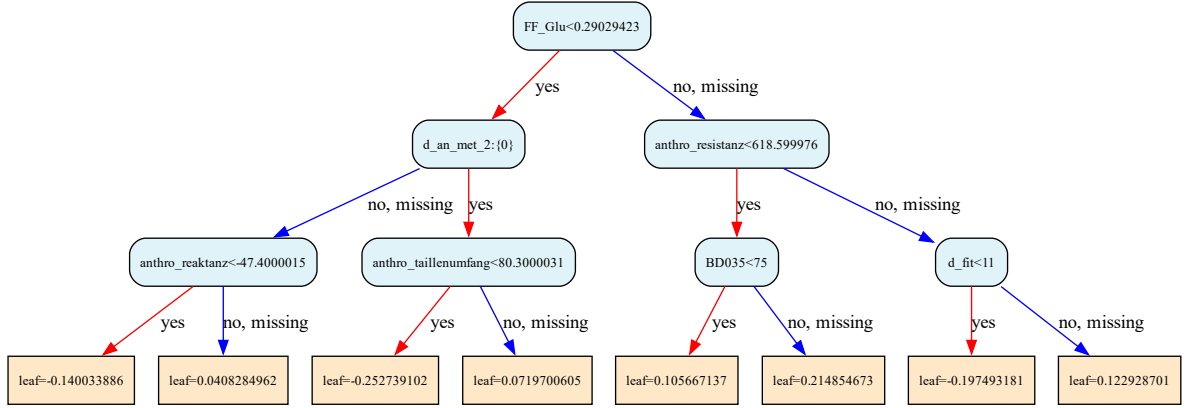


Figure 2.1.: Visualization of a CART tree from the XGBoost model used in the experiments. Rounded blue boxes are split nodes with their split criteria and the orange boxes represent the leaf nodes with their leaf scores.

To find the optimum set of functions f_k from the space F that fits the data given, we try to minimize a regularized objective function,

$$L(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad \text{where,} \quad \Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (2.4)$$

Here l is a differentiable loss function which is cross entropy loss that measures the error between the predicted labels $\hat{y}_i$ and actual labels y_i and $\Omega(f)$ is the regularization term that reduces the risk of overfitting by penalizing complex models.

It is computationally infeasible to minimize the objective function (2.4) across all the CARTs at once. Instead XGBoost uses an iterative approach where a new tree is added at each step such that it focuses on correcting the mistakes made by the previous trees thus minimizing the overall loss function. At step t loss function can be written as,

$$L^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t)$$

Here f_t is the tree added at the current step. XGBoost employs Newton's method by applying a second-order Taylor expansion to approximate the loss function at each data point x_i ,

$$L^{(t)} \approx \sum_{i=1}^n [l(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t)$$

$$g_i = \partial_{\hat{y}_i^{(t-1)}} l(y_i, \hat{y}_i^{(t-1)}) \quad , \quad h_i = \partial_{\hat{y}_i^{(t-1)}}^2 l(y_i, \hat{y}_i^{(t-1)})$$

are the first and second order derivatives of the loss function with respect to the previous predicted value. Substituting the values of $\Omega(f_i)$ and ignoring the constant terms,

$$\tilde{L}^{(t)} = \sum_{i=1}^n [g_i f_i(x_i) + \frac{1}{2} h_i f_i^2(x_i)] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

Grouping the data by leaf nodes,

$$\tilde{L}^{(t)} = \sum_{j=1}^T \left[\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) w_j^2 \right] + \gamma T \quad (2.5)$$

where $I_j = \{i \mid q(x_i) = j\}$ is the set of samples assigned to leaf j . Solving the above quadratic equation gives the optimal weight w_j^* for each leaf. The corresponding minimized loss function by substituting the value for optimal weight w_j^* is given by,

$$\tilde{L}^{(t)}(q) = -\frac{1}{2} \sum_{j=1}^T \frac{(\sum_{i \in I_j} g_i)^2}{\sum_{i \in I_j} h_i + \lambda} + \gamma T \quad (2.6)$$

To decide the best feature for a split, we compute the gain, which measures the improvement in the above objective function when a split occurs.

$$\text{Gain} = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma \quad (2.7)$$

where I_L and I_R denote the sets of instances in the left and right child nodes after the split. The feature that yields the maximum gain is chosen for the split. A higher gain indicates a more informative feature, which directly ties into our use of gain-based feature importance.

2.3. Mutual Information Based Feature Ranking

Mutual Information estimates the amount of information that a feature provides about the target variable [29]. Let x_i represent a feature in a dataset and y be the target variable. If the target variable y is completely independent of the feature x_i , their mutual information is zero. As the dependency between the feature and target increases, the mutual information value also increases.

$$I(x_i ; y) = H(y) - H(y|x_i)$$

where $H(y)$ is the entropy of y representing the total uncertainty in y and $H(y|x_i)$ is the uncertainty remaining in y after knowing the feature x_i . Thus, mutual information quantifies how much knowing the feature reduces the uncertainty in the target.

In terms of feature ranking, mutual information is used to identify the input features that share the most information with the target variable. Features with higher mutual information scores are considered more important, as they reduce the uncertainty about the target variable to a greater extent. This method is effective in detecting complex dependencies that simple correlation measures might miss. However, mutual information alone does not account for minimizing redundancy among features, which can be addressed by methods such as those proposed in [29].

2.4. Feature Mask Based Feature Ranking

Recent advances in deep learning has aided the development of more robust and scalable deep feature selection algorithms [30, 31]. The increasing popularity of deep learning based feature selection method can be attributed to three key factors:

- the scalability to handle large-scale high-dimensional datasets efficiently, enabled by advancements in modern hardware and parallel computing.
- flexibility to work with heterogeneous data types like images, videos and speech.
- modular design of neural architectures that allows for easy customization to fit the specific requirements of different applications.

To begin with, we describe a method based on the Feature Mask module [32], which identifies the relevant input features using a learnable mask. This mask assigns importance score to each feature, guiding the downstream learning process. The feature mask is learned via batch wise attenuation (BWA) and feature mask normalization applied over the mini-batches of data. This approach falls under the category of embedded feature selection, where a subset of K features is selected from the original D features during training and directly optimizing for performance on the downstream supervised task. The overall framework shown in Figure 2.2 comprises of two modules:

- Feature Mask (FM) Module $f(\cdot)$
- Learning Network $g(\cdot)$

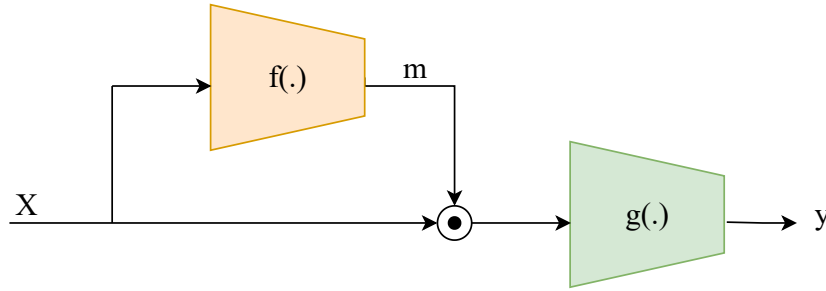


Figure 2.2.: General framework of feature selection method consisting of Feature Mask (FM) module $f(\cdot)$ and downstream learning network $g(\cdot)$. $f(\cdot)$ generates a feature mask m from input X . The learning network $g(\cdot)$ predicts the output y based on the masked input.

Let a dataset of N samples be denoted as $A = \{(x_1; y_1), (x_2; y_2), \dots, (x_N; y_N)\}$ with input samples $x_i = [x_{i,1}, x_{i,2}, \dots, x_{i,D}]^T \in R^D$ and corresponding ground truth labels y_i . During each training iteration, the module trains multiple mini-batches of data $X_B = [x_1, \dots, x_B]^T \in R^{B \times D}$ and outputs a normalized importance mask vector $m = [m_1, m_2, \dots, m_D]^T \in R^D$.

Each $m_i \in (0, 1)$ indicates the importance of i -th feature in the dataset. The training process involves learning an optimal feature mask $f(\cdot; m) : R^D \rightarrow R^D$ that maximizes the predictive performance of the downstream learning.

2.4.1. Building Blocks of Feature Selection Module

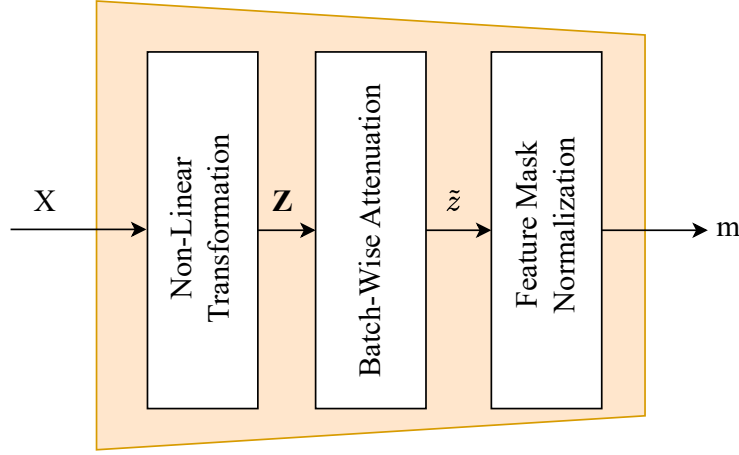


Figure 2.3.: Structure of the feature mask (FM) module $f(\cdot)$. FM module performs the mapping of raw input features to feature selection mask m

The feature mask module consists of three sub modules: i) nonlinear transformation; ii) batch-wise attenuation; iii) feature mask normalization shown in Figure 2.3 .

a. Non-Linear Transformation

The FM Module begins by applying non-linear transformation to capture the complex non-linear relationship among the raw input features. This transformation enables the model to assess the importance of features in combination with other features rather than in isolation. The transformation is composed of two steps: the input feature dimension R^D is transformed into a latent feature space R^E where $E < D$, thus removing certain irrelevant or redundant features in the process. Following this, the lower dimensional representations are mapped back to original input dimensions in order to compute weights for every individual feature.

Each mini batch $X_B = [x_1, \dots, x_B]^T \in R^{B \times D}$ is mapped to $Z = [z_1, \dots, z_B]^T \in R^{B \times D}$ using,

$$z_i = W_2 \cdot \phi(W_1 \cdot x_i + b_1) + b_2 \quad (2.8)$$

Here, ϕ is a non-linear function, $W_1 \in R^{E \times D}$ and $b_1 \in R^E$ are learnable parameters. In this work, $\tanh(\cdot)$ was chosen as the non-linear function due to its ability to retain both positive and negative values and thus can preserve important information.

b. Batch Wise Attenuation

At the end of the previous step, each input sample $x_i \in X^B$ is mapped to $z_i \in Z$. Since the non-linear transformation produces sample-specific representations it will not yield consistent representation of feature selection across the entire dataset. Therefore, this step averages

Z over the entire mini batch thus attenuating sample-specific variability and enforcing consistency across the batch. For each minibatch, we calculate

$$\tilde{z} = \frac{1}{B} \sum_{i=1}^B z_i \quad (2.9)$$

c. Feature Mask Normalization

The resulting $\tilde{z} = [\tilde{z}_1, \tilde{z}_2, \tilde{z}_3, \dots, \tilde{z}_D]^T$ is normalized using $\text{softmax}(\cdot)$ function to obtain the relative importance scores of input features.

$$m = \text{softmax}(\tilde{z}), \text{ where } m_i = \frac{e^{\tilde{z}_i}}{\sum_{j=1}^D e^{\tilde{z}_j}} \quad (2.10)$$

This ensures that all $m_i \in (0, 1)$ and sum to 1, making the importance scores interpretable and stable across training. All the three equations 2.8, 2.9, 2.10 that form the core of FM module are differentiable, thus allowing end-to-end training with the downstream model using gradient-based optimization.

2.4.2. Learning Network

The downstream learning Network $g(\cdot)$ can be any arbitrary neural network which is built specific to its downstream task which might be classification or regression.

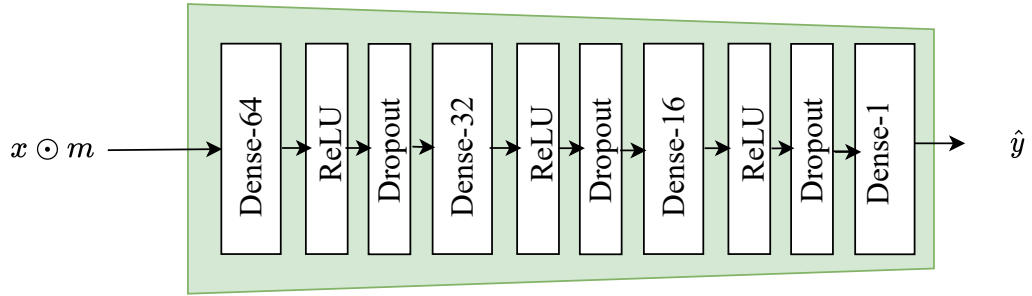


Figure 2.4.: Structure of the downstream learning network $g(\cdot)$. This fully connected network learns to predict the output label based on masked input features.

In this work, the learning network is a neural network classifier parametrized by θ_g . Each input sample is masked by element-wise multiplication with m :

$$x'_i = m \odot x_i$$

and is passed through $g(\cdot)$ to make predictions. The entire network is trained by minimizing the cross-entropy loss:

$$\min_{\theta_{FM}, \theta_g} \mathbb{E}_{(x,y) \sim p_{data}} \left[L(g(x \odot f(x; \theta_{FM}); \theta_g), y) \right] \quad (2.11)$$

After training, the FM module is applied over the entire dataset to compute a global feature mask m , from which the top- K most important features are selected. This method was chosen because of its simplicity, interpretability, and modularity. The model did not require additional regularization schemes or complex hyperparameter tuning.

2.5. TabNet Feature Importance

In this section, TabNet architecture [33] is described and how it integrates the soft feature selection ¹ in its every step to learn accurate and interpretable output mappings. TabNet shares conceptual similarities with the feature importance models discussed earlier in the following ways:

TabNet mimics the behavior of decision trees using conventional neural network components such as fully connected layers and activation functions to build a decision tree like structure. At each decision step, it performs feature selection, similar to the split selection in decision trees, but enhances this process through non-linear transformations of the selected features.

With respect to the Feature Mask (FM) module, TabNet model shares a similarity as both approaches learn a multiplicative feature mask that modulates the inputs during training. These soft masks are optimized jointly with the downstream task, aiming to highlight important features and suppress irrelevant ones. Therefore, comparing FM module & TabNet methods can offer insights on how different masking strategies impact the feature selection.

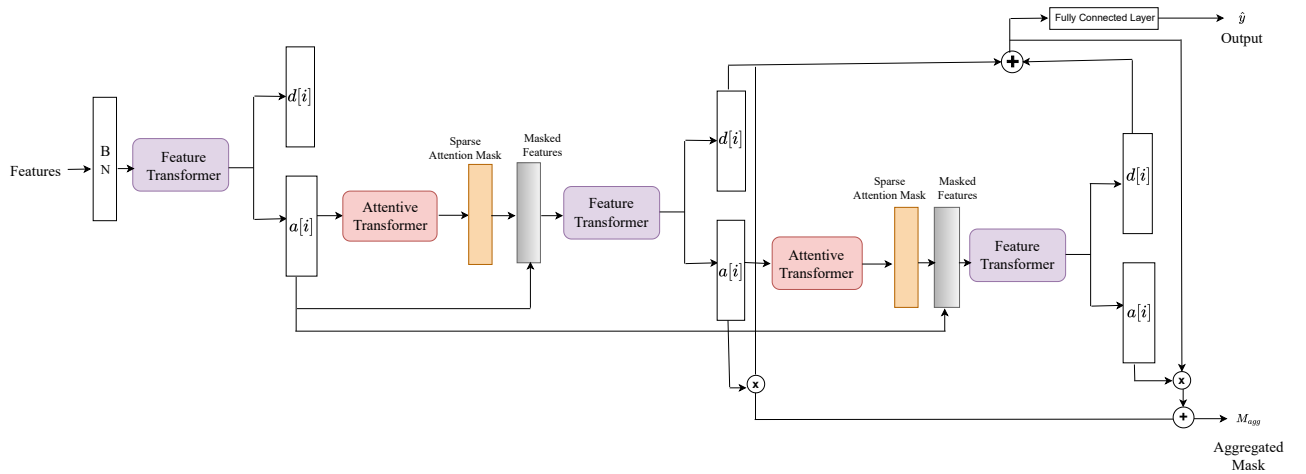


Figure 2.5.: Overall Structure of the 2 -step TabNet architecture. $d[i]$ and $a[i]$ are outputs of feature transformer at step i . $d[i]$ contributes to overall output. $a[i]$ is input to the subsequent attentive transformer. BN is the initial batch normalization layer.

TabNet is a multi-step neural network architecture as shown in Figure 2.5, composed of the following elements at each decision step:

1. Feature transformer block that maps the input features into more useful representations.

¹Soft feature selection refers to a technique where the algorithm assigns weights to the features indicating its importance rather than hard binary selection (inclusion or exclusion).

2. Attentive transformer processes the improved feature representation from the previous step and learns a feature mask, guiding the subsequent feature selection process.

The output from each feature transformer block is aggregated for final prediction and the masks from each attentive transformer is combined to derive the final feature importance scores.

2.5.1. Feature Transformer Block

Numerical input features are fed to the model directly while the categorical features are mapped to trainable embeddings, allowing the model to learn optimal representations at each step of the training. The feature transformer block is a n -layer network, where m layers are shared across all decision steps and the remaining $n-m$ layers are dependent on each decision step. Each layer consists of fully-connected layer, Ghost Batch Normalization (GBN) and Gated Linear Unit (GLU), a non-linear activation layer stacked sequentially.

The shared layers reuse the parameters (weights and biases) across different decision steps in the network. This reduces the number of trainable parameters ensuring robust learning and improved generalization. The output of the block is split using a split block into

- $a[i] \in R^{B \times N_a}$ to be used by attentive transformer of the subsequent step
- $d[i] \in R^{B \times N_d}$ which contributes to the overall output decision.

Formally, $[d[i], a[i]] = f_i(M[i] \cdot f)$, where $f \in R^{B \times D}$ is the input feature matrix and where $M[i]$ is the feature mask from the attentive transformer at step i .

Ghost Batch Normalization (GBN) [34] is applied in all normalization steps except the initial layer applied to the input features. GBN mitigates generalization issues with large batch sizes by virtually splitting each batch into smaller sub-batches of size B_V . Standard batch normalization is then applied to each sub-batch individually. The mean and standard deviation computed from each sub-batch are used to update the global running statistics (mean and variance) using an exponential moving average with momentum m_B . These aggregated statistics are treated as the statistics for the entire batch, effectively simulating the behavior of small-batch training even when using large batches.

Gated Linear Unit (GLU)[35] follows each normalization step. GLU computes the element-wise product of two linear transformations. One of the transformation is applied a sigmoid activation function which serves as a gate and the other is kept linear. The gating mechanism controls the flow of information allowing the model to dynamically suppress or retain features. The output is then passed to the next layer for further processing and is also added to the output of the next layer via a residual connection. This kind of skip connection allows the network to preserve information across decision steps and also helps stabilize the training by preventing vanishing gradients.

The overall output is computed as:

$$d_{out} = \sum_{i=1}^{N_{steps}} ReLU(d[i])$$

Finally, a softmax function is applied to a linear transformation of d_{out} to obtain final discrete predictions.

$$\hat{y} = \text{softmax}(W_{final} \cdot d_{out})$$

2.5.2. Attentive Transformer Block

Attentive transformer serves as the feature selection module in TabNet. It employs an attention mechanism that introduces sparsity into the feature set to learn the most relevant features at each decision step and passes it on to the feature transformer of the same decision step. The feature mask at i -th step $M[i] \in R^{B \times D}$ is calculated by,

$$M[i] = \text{sparsemax}(P[i-1] \cdot h_i(a[i-1]))$$

where $a[i-1]$ is the feature representation from the previous step and h_i is a trainable function given by the fully connected layer followed by GBN that forms a part of the attentive transformer block. $P[i]$ is the prior scales term that determines how frequently a feature can be selected by the model based on its usage in previous steps.

$$P[i] = \prod_{j=1}^i (\gamma - M[j])$$

where γ is the relaxation factor, $M[j]$ is the sparsemax activations (feature mask) from previous iterations. Intuitively, if a feature was used in the previous iterations, the model will pay more attention to the remaining features to reduce overfitting.

For instance, with $\gamma = 1$, a feature with large activation (e.g. 0.9) receives small prior scales ($1-0.9=0.1$) making it less likely to be selected. For further sparsity of selected features, the entropy of activations scaled by the hyperparameter λ gets added to the overall loss function.

The aggregated normalized feature mask is given by,

$$M_{agg-b,j} = \frac{\sum_{i=1}^{N_{steps}} \eta_b[i] M_{b,j}[i]}{\sum_{j=1}^D \sum_{i=1}^{N_{steps}} \eta_b[i] M_{b,j}[i]}$$

where $M_{b,j}[i]$ corresponds to contribution of j -th feature to the b -th sample at step i and $\eta_b[i]$ is the contribution score of i -th step to the overall mask.

3. Methodology

3.1. Datasets

The evaluation of feature ranking methods to identify the risk factors associated with diabetes was performed on two medical datasets, Behavioral Risk Factor Surveillance System 2015 (BRFSS)[36] and the German National Cohort (NAKO Gesundheitsstudie). Both datasets consist of information collected through interviews, survey responses and comprehensive biomedical examinations (e.g., blood pressure measurements, electrocardiograms, oral glucose tolerance tests). These diverse data sources were gathered to support early disease detection and risk factor identification, with the goal of enhancing preventive healthcare measures [37]. Initial exploratory experiments were conducted on a cleaned and reduced version of the BRFSS dataset, which contained fewer features and samples compared to the original dataset of over 330 features. These experiments focused on identifying risk factors for diabetes, allowing for comparison with established findings from the literature. They served as a benchmark to assess the effectiveness of various feature ranking and selection techniques. The insights gained from the preliminary results laid the foundation for subsequent, more complex analyses using the NAKO dataset.

Table 3.1.: Overview of datasets

Dataset	#Features	#Samples	#Classes	Imbalanced Dataset
BRFSS2015	21	70692	2	No
NAKO	149	13669	2	Yes (85:15, Non-Diabetic Vs Diabetic)

Table 3.1 summarizes the details of the datasets used in the experiments. Both datasets contain a mix of categorical and continuous features. The BRFSS dataset was already cleaned and did not require much preprocessing but NAKO dataset was unprocessed and required extensive preprocessing. The detailed steps involved in preparing the NAKO dataset for analysis are described in the following section.

3.2. Data Preprocessing

The NAKO dataset was initially augmented with additional files containing Magnetic Resonance Imaging (MRI) -derived measurements, including volume, maximum axial cross-sectional area, and fat fraction values for eight muscle groups namely, the left and right gluteus and psoas muscle groups, as well as the left and right thigh extensors and adductors.

These measurements were computed using the predicted segmentation masks of approximately the first 10,000 patients. The datasets were merged based on the patient IDs, and only records with available MRI results were retained, resulting in a final dataset containing 12023 samples.

3.2.1. Missing Values Imputation

The missing values which were either left unanswered or encoded with placeholder values such as 999, 8888, 9999 were identified from the metadata documentation provided with the dataset. Columns with more than 75% missing values were removed to avoid introducing excessive noise due to a high proportion of imputed data. Figure 3.1 shows the distribution of features based on the percentage of missing values. The missing data in the remaining columns were imputed with k-Nearest Neighbors (kNN) imputer from the scikit-learn library. It was observed that imputation was especially important for neural network-based feature selection methods, as the presence of missing values hindered effective learning and the identification of relevant features.

kNN imputation [38] fills the missing values in a dataset by leveraging the k-Nearest Neighbors algorithm. It involves finding the k most similar data points (nearest neighbors) based on the available features and calculating the average of their corresponding values for the missing attribute. This approach preserves the underlying relationships between features, better than simpler methods such as mean or median imputation, often resulting in improved model performance. For NAKO dataset, kNN imputation was performed considering the 2 nearest neighbors for estimating the missing entries.

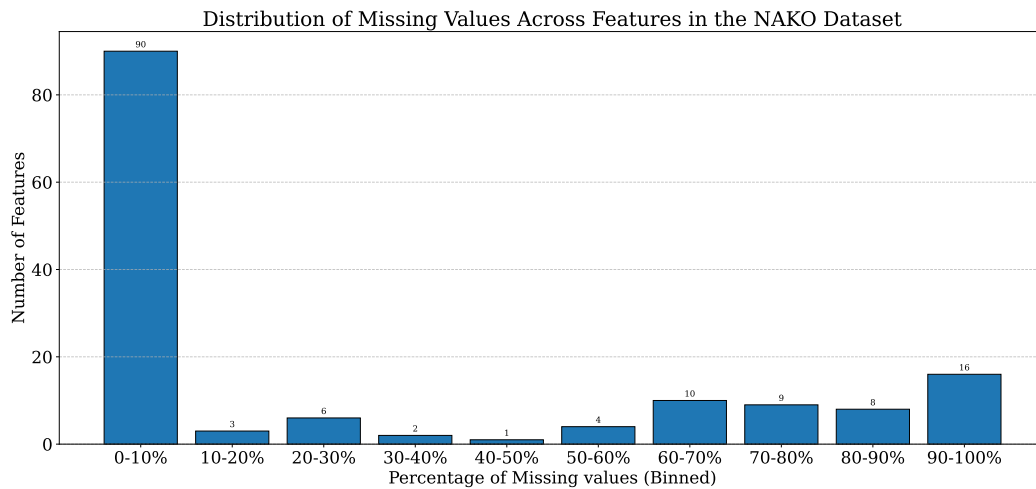


Figure 3.1.: Distribution of missing data percentages across features in the NAKO dataset. The x-axis represents bins of missing value percentages and y-axis indicates the number of features that fall within each bin.

3.2.2. Feature Engineering

Initial Feature engineering was performed to transform the raw data into meaningful input features and to improve the performance of machine learning models. In NAKO dataset, feature engineering was performed to reduce redundancy and lower dimensionality resulting in improved data quality and preventing overfitting. Several features comprising of two or three related measurements such as hand strength and ultrasound scan results were aggregated to enhance data consistency and reduce dimensionality. In the NAKO dataset, a significant degree of redundancy was observed among variables associated with physical activity and the exertion it induced. To address this, features exhibiting high correlation and conceptual overlap were systematically consolidated into single representative variables, and the original raw features were removed. For example, hand strength was measured three times for each hand, resulting in six separate features. These were averaged to form a single composite feature representing hand strength for each hand. A similar approach was applied to abdominal ultrasound scans, where paired measurements were combined into unified features. This aggregation was guided by the degree of agreement among the original measurements, ensuring that the combined features retained the relevant information while minimizing redundancy and potential noise in the dataset. For the MRI scan results of muscle groups, the volumes and cross-sectional areas of the left and right sides were summed to obtain total values for each muscle group, while fat fractions were averaged to reflect the mean fat content across both sides. For the features which dealt with the ordinal data on the duration of physical activity, related columns were consolidated based on mode of their values. This approach was chosen because participants typically responded mostly to one or two these related columns leading to sparse responses across these columns. Derived features were also created, such as Body Mass Index (BMI), which was calculated from height (*anthro_groe*) and weight measurements(*anthro_gew*) of the patients based on the formula in [39]. The binary categorical input variables, originally encoded as 1 and 2, were remapped to 0 and 1, respectively to avoid implying any inherent ordinal relationship during model training

The target feature was defined as a binary variable taking the value 1 if a person has diabetes and 0 otherwise. The value of the target column was primarily based on the column that indicated the patient's diabetes mellitus status (*d_an_met_1*). For patients with missing (*d_an_met_1*) values, *sa_hba1c* measurements were considered. As per [40], patients with *sa_hba1c* ≥ 48.0 mmol/mol were labeled as diabetic and were assigned value 1, while those with *sa_hba1c* < 48.0 mmol/mol were labeled as non-diabetic and were assigned value 0. Additionally, any patients initially reported as non-diabetic in *d_an_met_1* but with *sa_hba1c* ≥ 48.0 mmol/mol were reclassified as diabetic with target value 1. This approach ensured consistent and comprehensive labeling of the diabetes status.

Further, features that were related to diabetes measurement methods like *sa_ogtt0*, *sa_ogtt2*, *ogtt_o2* and *ogtt_o3* which indicate if a patient is treated for diabetes were removed. These features most directly indicated presence of diabetes however our aim was to identify other important factors that might led to the diabetes. Although inclusion of these features improved prediction accuracy and they ranked highly in feature importance during experiments conducted as a sanity check, these results were not included in the thesis as they primarily reflect direct diagnostic indicators rather than contributing factors.

¹Descriptions of all feature names are available in the Appendix section.

3.2.3. Handling Imbalance

It is very common for medical dataset to have imbalance, NAKO dataset also exhibits class imbalance of ratio 85:15 between patients without diabetes and patients with diabetes respectively. In order for the model to not be biased towards the majority class, the imbalance in the dataset needs to be addressed for good generalization. However, since the primary focus of this work is on feature importance ranking, synthetic data augmentation techniques like Synthetic Minority Over-sampling Technique (SMOTE) [41] or Adaptive Synthetic sampling (ADASYN) [42] were avoided. These methods might introduce unrealistic synthetic patient records that could distort the underlying feature distributions and consequently bias the feature ranking results. Instead we opted to use class weighting, which adjusts the loss function by assigning weights and penalizing errors on minority class more heavily. This approach encourages the model to pay more attention to underrepresented samples without altering the original data distribution.

3.3. Feature Ranking & Selection Methods

Various feature ranking methods were employed to identify the most influential subsets of features and their ranking. The methods include XGBoost feature importance, the Feature Mask (FM) module based selection, TabNet feature importance, and Mutual Information (MI) based feature importance. An overview of each of these methods was provided in Chapter 2. These methods were selected to study the comparisons between model agnostic, statistically driven methods (MI feature importance) and model based methods. The performance of deep learning based feature selection methods (FM module and TabNet) relative to boosted tree models, which are among the most widely used models for tabular data were also examined. The features are ranked according to their importance scores produced by each method.

3.4. Downstream Classifiers

One of the main challenges in feature selection is that the ground truth of optimal feature subsets are unknown. This makes it inherently difficult to assess the quality of selected features, because an exhaustive search over all possible feature subsets is computationally infeasible. Unlike domains such as computer vision, where selected features in the form of image pixels can be visualized for intuitive understanding, tabular datasets lack such interpretability. As a result, a practical approach to evaluate the effectiveness of selected features is to assess their impact on downstream classification performance.

In this study, the selected features are assessed by measuring the balanced accuracy achieved by four different classifiers: Logistic Regression, CatBoost, Multi-Layer Perceptron (MLP), and an Ensemble Classifier. The classifiers were chosen to represent a spectrum of learning paradigms from linear models to gradient boosted trees to deep neural networks. Each model is trained and evaluated on feature subsets of varying size, $k = 10\%, 30\%, 60\%$ and 100% of the total features. A brief description of each downstream classifier is detailed in the following subsections.

Logistic Regression (LR):

Logistic Regression [43] serves as a simple, interpretable, and widely used baseline classifier for supervised binary classification problems. It assumes a linear relationship between input features and the log-odds of the outcome, making it suitable to test how well the selected features linearly separate classes. In the experiments, the model is trained up to maximum of 100 iterations, with balanced class weights if necessary and L2 regularization is used to prevent overfitting.

CatBoost:

CatBoost [44] is a gradient boosting algorithm that shares core principles with XGBoost but optimized for handling categorical features and reducing overfitting. The model builds balanced decision trees where the same feature and the split condition are applied uniformly at each level of the tree. The algorithm employs ordered boosting and ordered target statistics, which ensure that the current data point is excluded when computing gradient updates and categorical encodings. These mechanisms prevent target leakage thus making CatBoost a reliable choice for evaluating the quality of feature ranking methods without being prone to overfitting.

Multi-Layer Perceptron (MLP):

MLPs [45] are capable of modeling complex, non-linear decision boundaries by stacking multiple layers of neurons followed by a non-linear activation function. The network learns by iteratively adjusting the weights of the neurons to minimize a predefined cost function using gradient-based optimization. In this study, a simple four layer architecture is employed, with Rectified Linear Unit (ReLU) activations in hidden layer and sigmoid activation in the output layer. MLP evaluates whether the selected features provide sufficient information for deep learning models to learn intricate patterns, beyond simple linear separability.

Ensemble Classifier:

An ensemble classifier [46] employed in this study combines the predictions of all the above mentioned classifiers and determines the final output through hard voting (based on individual predicted labels). Including an ensemble approach helps assess whether the selected feature subset generalizes well across different model types and compensates for the weaknesses of individual classifiers.

The configurations of the downstream classifiers are kept consistent across all experiments to ensure fair comparison between different feature ranking methods and subsets. The goal is not to maximize the classification accuracy for each downstream classifier but to evaluate the predictive power of the selected features and analyze performance trends across different feature ranking methods as the number of selected features varies.

3.5. Performance Metrics

The entire feature selection and evaluation pipeline is validated on a combination of classification performance and ranking consistency metrics. To further ensure that the evaluation reflects true generalization and is not biased towards any particular subset of data, all experiments are cross validated over 5 independent runs with different random seed.

3.5.1. Balanced Accuracy

Balanced accuracy was employed as the primary evaluation metric for both feature selection methods and downstream classifiers. This metric is particularly appropriate for imbalanced datasets, as it accounts for performance on both the majority and minority classes. In binary classification tasks, it is defined as the average of recall for each class.

$$BalancedAccuracy = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right)^{\dagger}$$

It is particularly suited for medical datasets, where the minority class (i.e. high-risk patients) is often of greater interest but underrepresented. Balanced accuracy mitigates the bias towards the majority class by giving equal importance to both classes, thus providing a more reliable and informative measure of performance compared to standard accuracy.

3.5.2. Spearman Rank Correlation

To evaluate the agreement between different feature ranking methods, pairwise Spearman rank correlation is computed between the feature rankings produced by each method. Spearman correlation (ρ) measures the strength and direction of a monotonic relationship between the rank of a features from two different methods. The values ranges from $[-1, 1]$ where 1 indicates a perfect perfect agreement while, -1 indicates a complete inverse agreement. A heatmap of the Spearman correlation coefficients is generated to visualize the pairwise agreements among all feature ranking methods. The Spearmann correlation coefficient is given by:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where, d_i is difference between the two ranks of feature i ,
 n is the number of features

[†] TP - True Positive, TN - True Negative, FP- False Positive, FN - False Negative

3.6. Overview of Feature Ranking and Evaluation Pipeline

Figure 3.2 illustrates the typical pipeline of a feature ranking and evaluation algorithm. The workflow is designed to systematically process raw input data, apply various feature ranking techniques, and assess the effectiveness of the selected features using diverse set of classifiers.

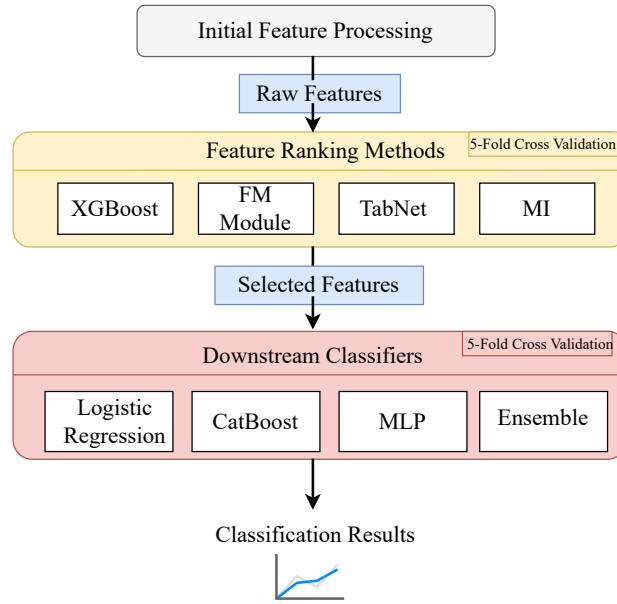


Figure 3.2.: Overview of feature ranking and evaluation pipeline

Initial Feature Processing

This step involves the preprocessing of input data to ensure its quality and consistency for effective model training. Operations such as handling missing values, standardizing the features and aggregating related features are performed based on dataset specific requirements. The preprocessing details for each dataset used in this study are provided in Section 3.2.2. The result of this step is referred to as raw features, which serves as the input for the feature ranking phase.

Feature Ranking

In this step four different feature ranking strategies are employed to learn the most relevant features for the classification task. A stratified ¹ 5-fold cross-validation is applied during feature ranking to ensure consistency of selected features across different training-validation splits. For each method, the feature importance scores or feature mask values depending on the algorithm are computed in each fold and then averaged to obtain the final feature importance ranking or selection mask. All feature ranking methods are applied only on the training dataset to prevent information leakage. The trained feature ranking is evaluated on

¹Preserves the original distribution of each class

a test set for iterative refinement such as adjusting the hyper parameters or trying different configurations to select the configuration that yields the best average performance. The top ranked features are then evaluated on downstream classifiers in the following step.

Downstream Classifiers

A set of downstream classifiers is used to evaluate the predictive power of feature subsets obtained from the feature ranking algorithms. It is important to note that some of the feature ranking methods are embedded methods, where their selection is inherently tied to a specific classifier. For example, the XGBoost-based feature importance is optimized for the XGBoost classifier, while the FM Module is designed to work in conjunction with its learning network, a multilayer perceptron in this case. To ensure a fair and unbiased evaluation, we apply a separate set of downstream classifiers that are independent of the models used during feature selection. These include a logistic regression model representing the linear models, a CatBoost model representative of gradient boosted decision trees and a multi layer perceptron model representative of applying deep neural networks to tabular dataset. Finally an ensemble model, combining the strengths of all the above classifiers to capture a wider range of decision boundaries. Each downstream classifier is trained on the selected features across varying feature subset sizes to analyze the impact of dimensionality on predictive performance. The classifiers are evaluated on a held-out test set, containing only the selected features, to generate predictions. To reduce variance in performance estimation, we apply stratified 5-fold cross-validation to evaluate each configuration. The primary metric is balanced accuracy, which accounts for class imbalance and is computed by averaging the metric across the five folds. The evaluation results are visualized through plots illustrating balanced accuracy versus the number of selected features. These visualizations provide a comprehensive overview of the effectiveness and efficiency of different feature selection methods in preserving relevant information for downstream predictive tasks. The next section presents the results of the experiments conducted to evaluate various feature ranking methods and their influence on the performance of downstream classifiers.

4. Results & Discussion

RQ.1 What are the most import features that can be considered as medical risk factors for the onset of diabetes?

Figure 4.1, 4.2, 4.3, 4.4 illustrates the feature importance plots derived from various methods, applied to the BRFSS dataset with a balanced class distribution. The most influential features consistently identified across all methods are High Blood Pressure (High BP), General Health (GenHlth), High Cholesterol (HighChol), Age, and Body Mass Index (BMI). It is observed that there is a significant overlap in the top influential features. This aligns well with clinical intuition and existing medical literature, where individuals with other metabolic conditions such high cholesterol and high blood pressure are more susceptible to diabetes [47, 48]. Similarly, age and BMI are widely recognized as common risk factors for the onset of diabetes [49]. Agreement among methods regarding the least important subset of features was limited. However, features such as access to any health care (AnyHealthcare) and avoiding a doctor visit due to cost (NoDocbcCost) were consistently ranked among low importance subsets. Features related to physical health (PhysHlth), difficulty in walking (DiffWalk), and heart-related diseases (HeartDiseaseorAttack) were generally assigned medium importance by most methods.

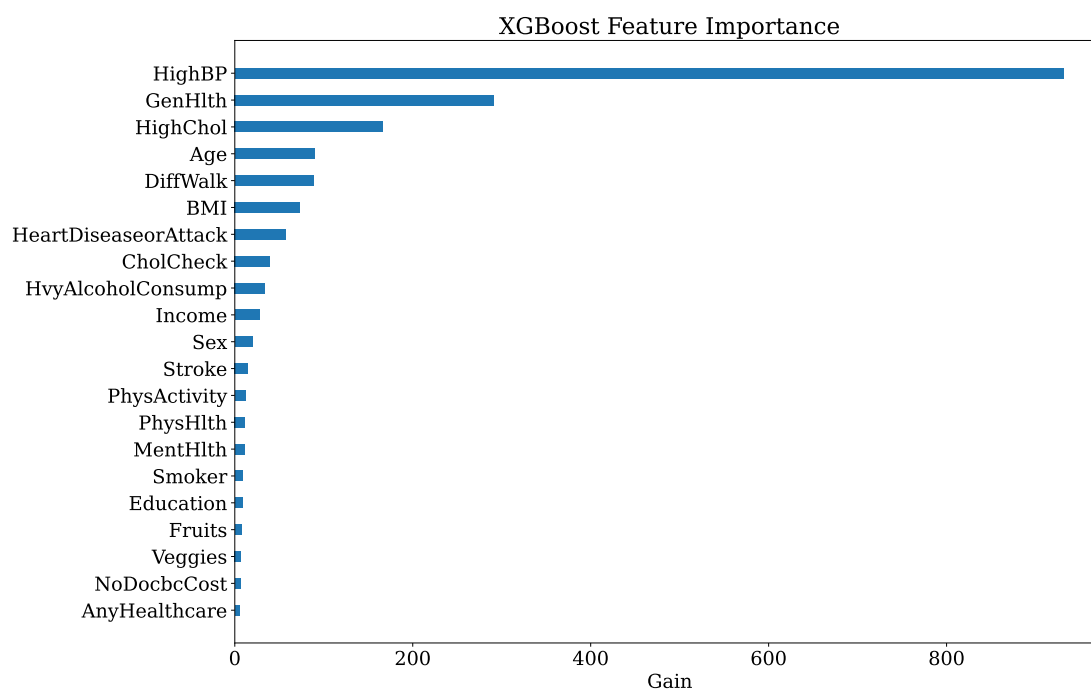


Figure 4.1.: Feature rankings based on the gain metric from XGBoost classifier on BRFSS dataset

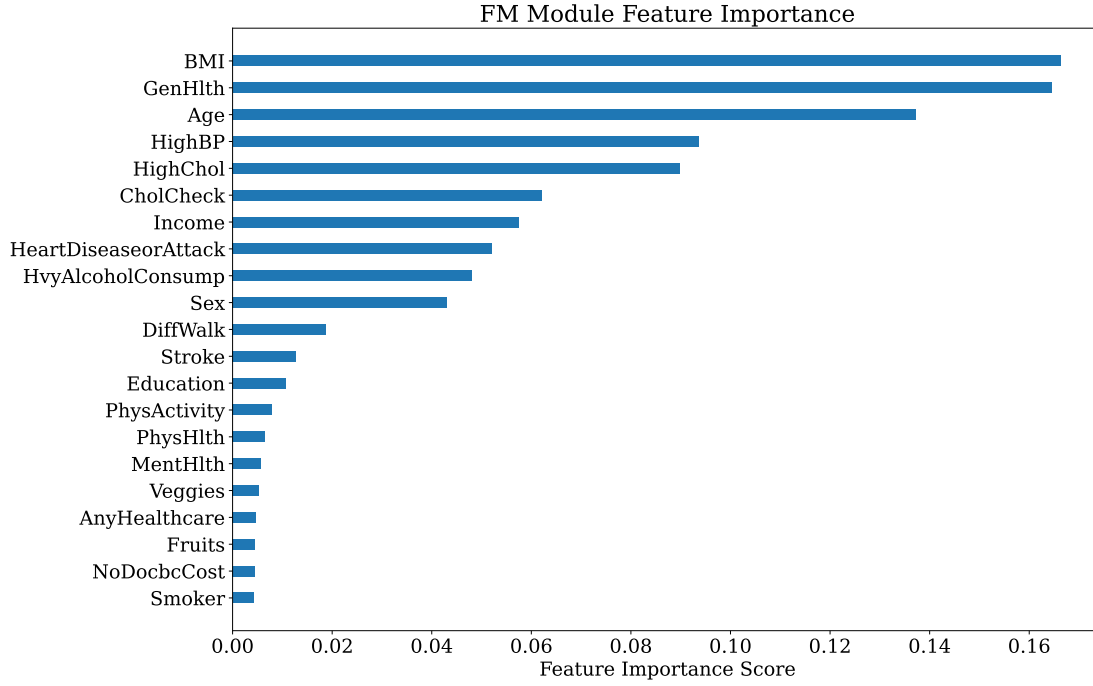


Figure 4.2.: Feature rankings based on the feature mask values (m) from FM Module on BRFSS dataset

The feature importance values from the TabNet model sheds light on features which were most frequently selected or attended to at the each decision step during the learning process.

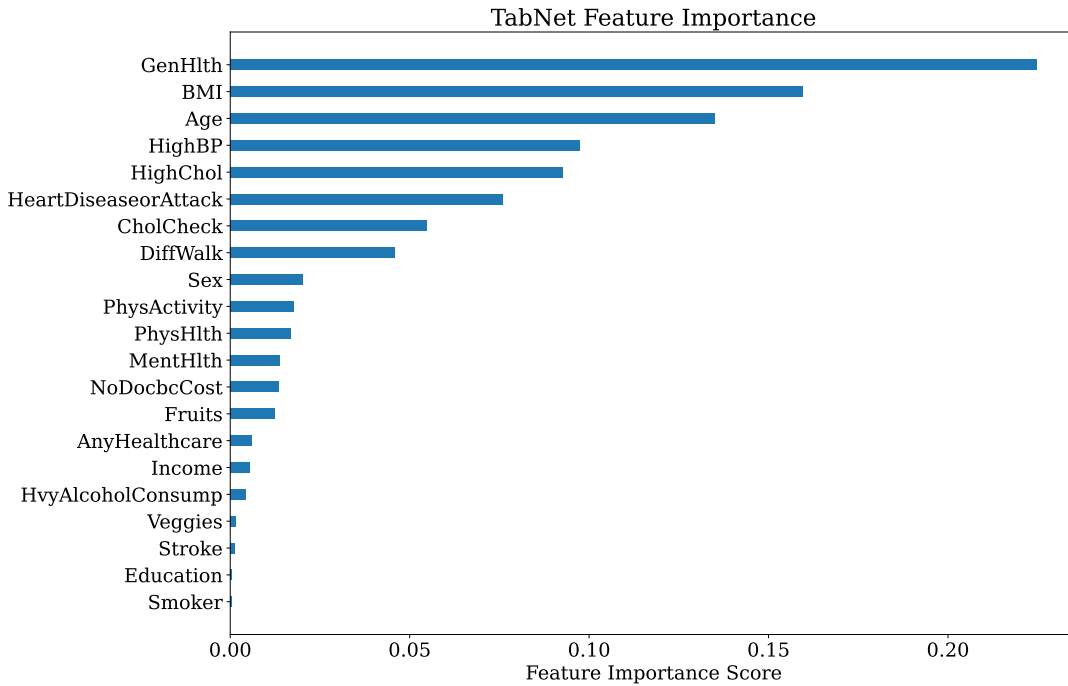


Figure 4.3.: Feature rankings based on TabNet aggregate feature importance mask M_{agg}

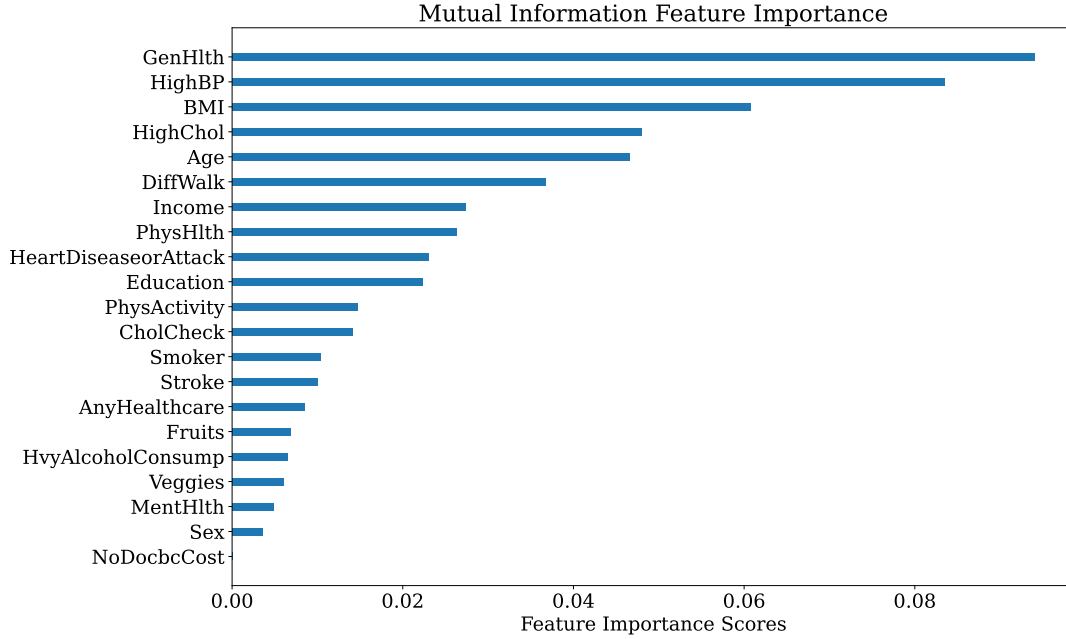


Figure 4.4.: Feature rankings based on mutual information scores on BRFSS dataset

From Figure 4.1 XGBoost appears to place a high importance on a single feature (High BP) while other feature ranking methods tend to distribute the importance across the features. This behavior can be attributed to the fact that High BP is likely used in the early splits across multiple decision trees. This is typical of tree based models such as XGBoost where features involved in early split tend to be favored as these features influence a larger subset of the data and contribute significantly to reducing the loss. Additionally, such methods might exhibit a bias toward features with higher cardinality or more potential split points.

To validate the model-based feature importance results, various statistical visualizations were employed to examine the distributions of highly influential features in relation to diabetes outcomes. The results from Figure A.1, A.2¹ showed that individuals with conditions like High Blood Pressure and Cholesterol had a higher prevalence of diabetes, thereby supporting the high feature importance scores assigned by the feature ranking methods. For continuous features like BMI, it was categorized further into underweight, normal, overweight and obese categories. The corresponding plot Figure A.3 effectively showed a strong association between higher BMI values and increased diabetes prevalence. The proportion of diabetic individuals being more dominant in obese classes supports the findings that higher BMI values is one of indicators of diabetes in the dataset. These visual analyses not only corroborated the results of feature importance rankings but also provided an intuitive understanding of the relationship between these predictors and the target outcome.

In order to assess the generalization capability of our feature ranking and evaluation framework, our experiments are extended to NAKO dataset which presents a more realistic, real-world scenario. After initial feature engineering, the dataset comprises of 84 features and

¹In Appendix

12,023 samples. However, a major challenge with this dataset was the presence of significant class imbalance where, samples with disease representing only about 15% of the total samples (approximately an 85:15 class ratio). The limited number of samples compared to the high dimensionality introduces a classical HDLSS (High-Dimensional Low Sample Size) problem [50], which challenges the robustness of feature ranking and classifier performance.

Out of 84 features, 25 most influential features were identified by various feature importance methods are presented in Figure 4.5, 4.6, 4.7, 4.8 and these features form the basis of the following discussion. High blood pressure (d_an_cv_6), elevated blood lipids, cholesterol, or triglycerides (d_an_met_2), age (basis_age) and body mass index (BMI) consistently emerged as top-ranked features. This pattern mirrors the results obtained from the BRFSS dataset, where the similar set of features demonstrated high importance across multiple selection techniques. Such consistency across two distinct datasets supports the reliability of the feature ranking methods employed, irrespective of variations in dataset characteristics. Furthermore, waist circumference (anthro_taillenumfang) and BMI which are among most important features are also the clinically studied risk factors for type 2 diabetes [51].

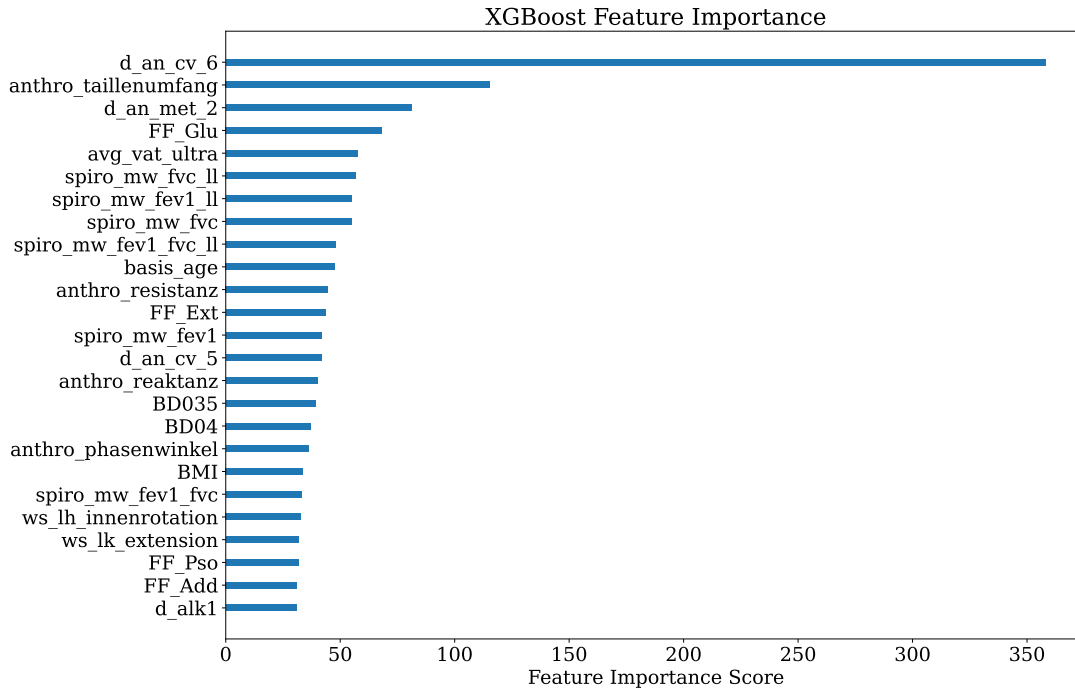


Figure 4.5.: Feature rankings based on the gain metric from XGBoost classifier on NAKO dataset

Additionally, features related to fat fractions in specific muscle groups were also consistently selected. Notably, fat fraction in the gluteus muscle group (FF_Glu) appeared across all four feature selection methods. Similarly, fat fractions and the maximum axial cross sectional area from other muscle groups such as extensors, psoas and adductors (FF_Add, FF_Ext, FF_Pso, CSA_Glu) were consistently selected among top 25 features by most methods. This suggests that localized fat distribution may serve as a physiologically meaningful predictor of diabetes risk. This interpretation is further supported by existing studies conducted on

obese individuals, which highlight the association between regional fat accumulation and insulin resistance or metabolic dysfunction [52]. It is worth noting that most of the features consistently ranked at the top were related to fat distribution. Features such as `avg_vat_ultra`, `anthro_reaktanz` and `anthro_resistenz` were ranked among the most influential features by all methods. `avg_vat_ultra` measures abdominal fat accumulation, `anthro_reaktanz` and `anthro_resistenz` are results of bio impedance analysis, which measures the total body fat. Such evidence reinforces the potential of fat distribution metrics as novel biomarkers in diabetes prediction frameworks.

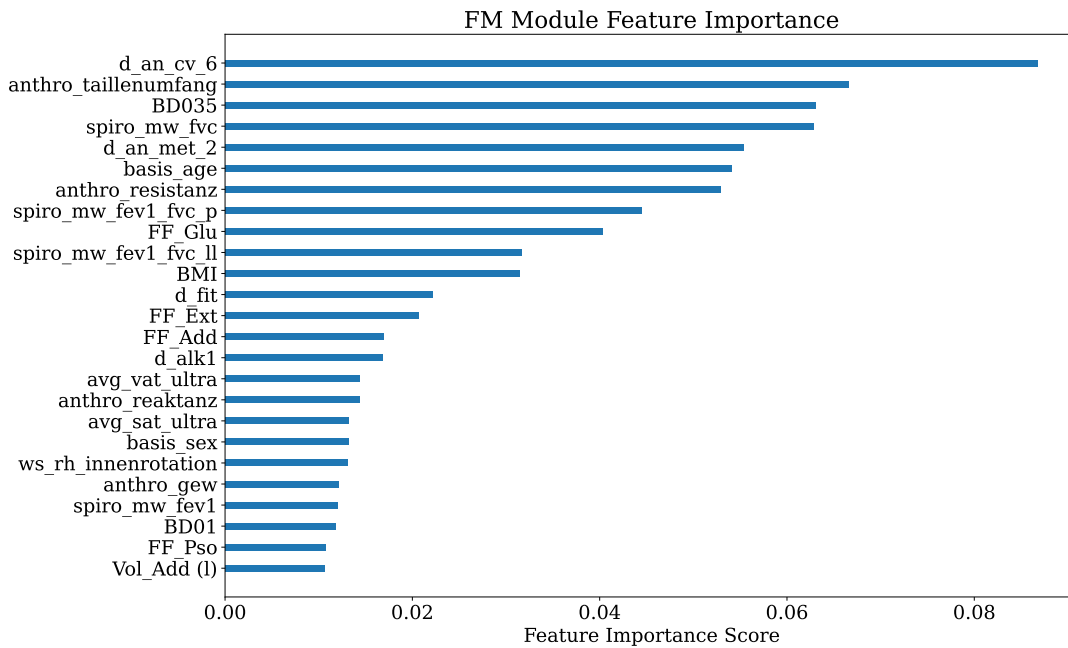


Figure 4.6.: Feature rankings based on the feature mask values (m) from FM Module on NAKO dataset

Furthermore, several features indicative of a person's objectively measured fitness parameters or functional capability, such as various spirometry values measuring the forced expiratory vital capacity (`spiro_mw_fvc`, `spiro_mw_fev1_fvc_p`) were among the important features in most methods. Interestingly, other fitness indicators such as range of motion in knee (`ws_rk_extension`, `ws_lk_extension`) and hand rotation (`ws_rh_innenrotation`) also appeared as key features, although different methods selected different subsets of these features. Self reported fitness levels such as `d_qol3_1`, `d_fit` were also among the top influential variables. Their presence suggests that both measured and self reported fitness levels may play a significant role in diabetes prediction, adding an additional behavioral and lifestyle dimension to the risk profile which has been discussed in numerous studies [53].

¹Descriptions of all feature names are available in the Appendix section.

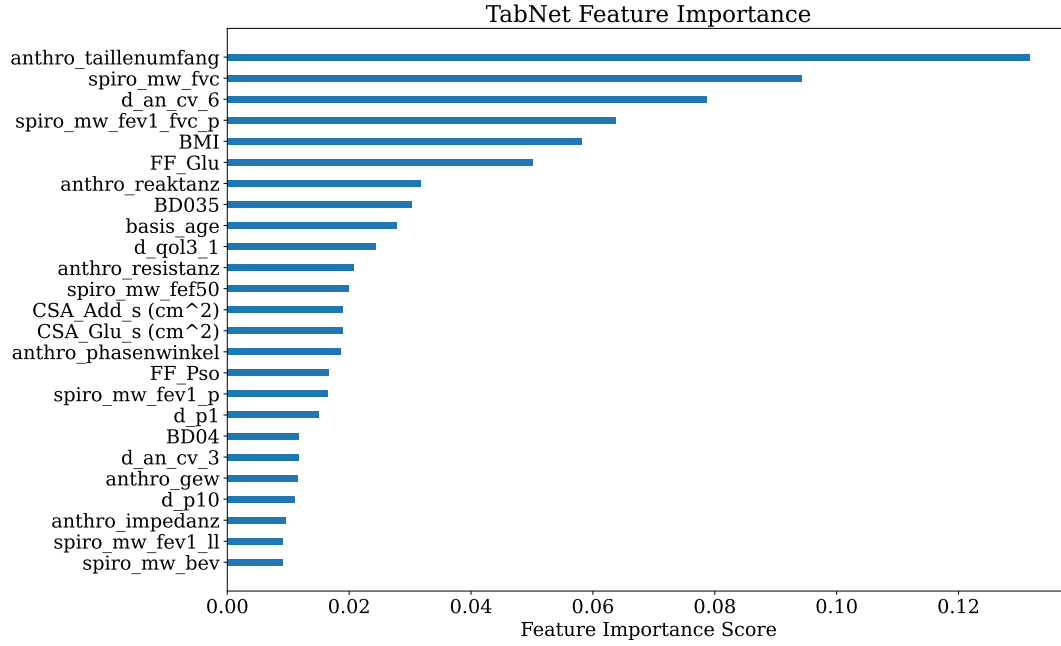


Figure 4.7.: Feature rankings based on TabNet aggregate feature importance mask M_{agg} on NAKO dataset

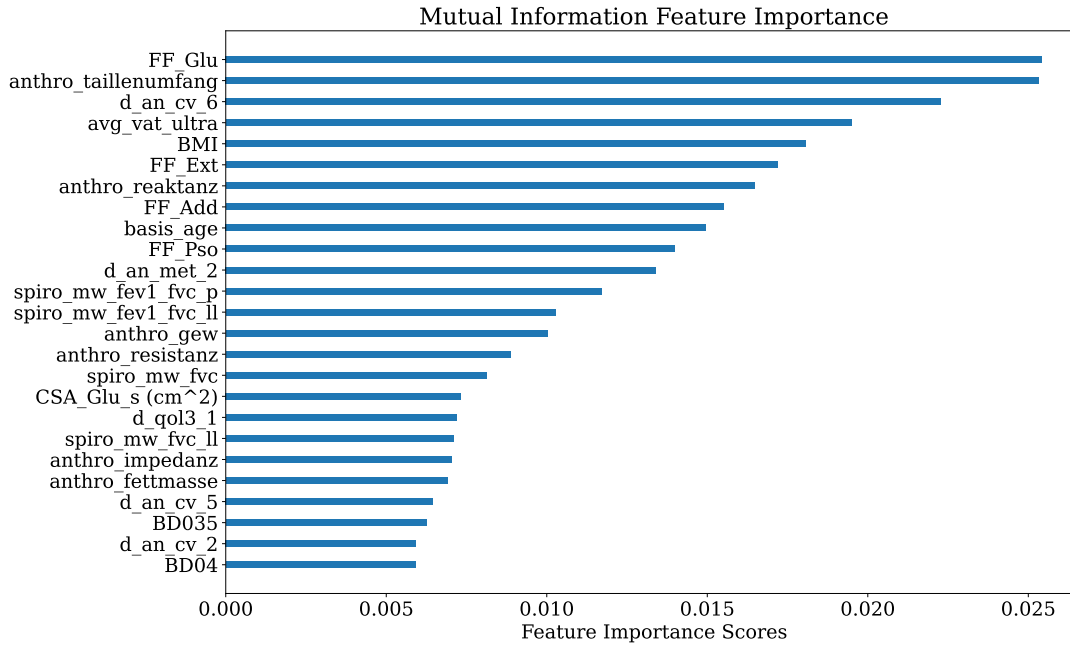


Figure 4.8.: Feature rankings based on mutual information scores on NAKO dataset

Feature ranking methods highlight the features that helps the model perform better (discrim-

inates the target labels better) and does not account for redundancy in the features. It can be noted that most feature ranking methods discussed here featured both BMI and weight (anthro_gew) among important features even though BMI is a derived feature from weight and may contain overlapping information. However techniques such as minimum Redundancy Maximum Relevance (mRMR) and regularization based techniques like Lasso can be used in feature selection to balance between relevance and redundancy.

4.1. Evaluation of top-K features

RQ.2 How does the top ranked features impact the performance of the downstream model?

The balanced accuracies obtained for various subsets of top ranked features are evaluated on four downstream classifiers: Logistic Regression, CatBoost, MultiLayer Perceptron (MLP) and Ensemble classifier for BRFSS dataset are shown in the Figure 4.9.

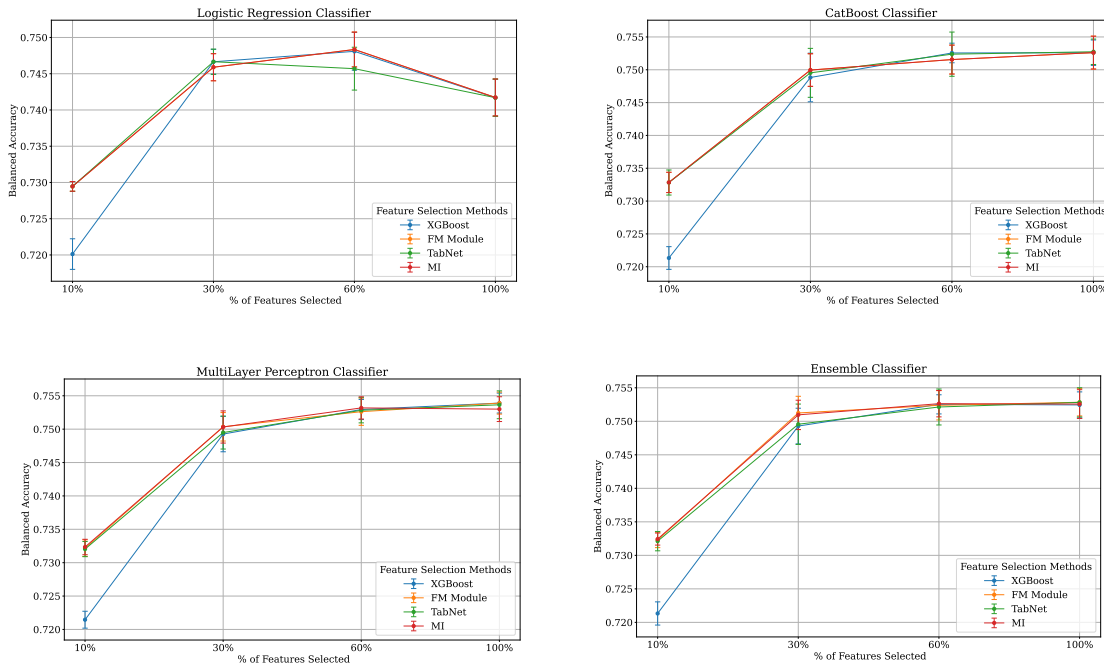


Figure 4.9.: Plot of balanced accuracy obtained for various percentages of top ranked features in BRFSS dataset. Results are visualized for logistic regression, CatBoost classifier, multi layer perceptron and ensemble classifier respectively.

In general, it is observed that the accuracy of the downstream classifiers increased as more informative features were added. There is a relatively steep increase in the accuracy in the initial stages between 10% to 30%. Beyond that, the accuracy either improves marginally or plateaus suggesting that all informative features are being selected initially. This results in diminishing returns from additional features in the later stages. An interesting observation with logistic regression was observed, where the model reached the highest accuracy with 13 features (60% of total features) and a slight decline in accuracy was observed when all available features were included. This drop in performance can be attributed to overfitting, where

a model captures not only the underlying patterns in the data but also random fluctuations or noise specific to the training dataset. This can be attributed to curse of dimensionality and multicollinearity [50], as the number of features increases, the feature space becomes increasingly sparse and large data is required to generalize effectively.

However, in CatBoost the improvement in performance is almost negligible with the addition of more features. This is due to its built-in regularization mechanisms and gradient boosting framework, which mitigates overfitting. So the final addition of less informative features have almost no effect on the performance improvement. Similar trends can be seen with both MLP and ensemble based models where there was a steep improvement within 30% of features and after that the performance improvement almost plateaus. These models also demonstrated resilience to overfitting due to their capacity for regularization. In the case of TabNet, their inbuilt attention mechanism promotes selective feature usage. With too few features, almost all feature selection methods have similar feature importance rankings. XGBoost had the least performance at 10% of the total features in all downstream classifier performance. XGBoost ranked categorical variables like High BP, GenHlth and HighChol as most important. Age and BMI which were among top features in other feature ranking methods were placed little lower in XGBoost. This could explain the drop in the performance. The maximum achieved accuracy on test set during feature selection was 0.753. Similar accuracy values ranging from 0.750 to 0.753 were achieved with most downstream classifiers with just top 7 features.

The performance of downstream classifiers on selected features from the NAKO dataset is presented in Figure. 4.10, 4.11, 4.12, 4.13. In general a trend can be observed across classifiers where the maximum accuracy is often achieved with only 30% of the available features. This supports the validity of the feature importance rankings, indicating that a relatively small subset consisting of highly informative features can drive strong predictive performance. Notably, few classifiers achieved competitive results even with only 10% of the total feature set (top 8 features out of 84).

This outcome also indicates that the NAKO dataset contains few highly informative features that dominates the learning process. The inclusion of noisy or irrelevant features do not contribute to increase in the performance and might even result in a decline in the performance. This can be evidently observed in Figure 4.10 where the model achieves its highest accuracy of 0.768 (76.8%) with only the top 8 features. As a linear model, logistic regression displays a clear preference for smaller feature subsets. Among the feature ranking methods, features selected using TabNet yielded the best performance, closely followed by feature from FM Module and XGBoost ranking methods. As more features were added, a decline in the model's performance was observed.

A similar pattern emerged with the CatBoost classifier where the model reached peak performance at 30% of total features (top 25 features) with accuracy of about 0.780 (78%). Top feature subsets derived from XGBoost and the FM Module performed best for this classifier. The strong performance of CatBoost when using feature subsets ranked by XGBoost suggests that feature importance methods which share the same underlying principles with the learning algorithm may be more effective. Since both models are based on gradient boosted decision trees, they may prioritize similar feature patterns such as handling of non-linear interactions and considering importance of splits leading to complementary behavior.

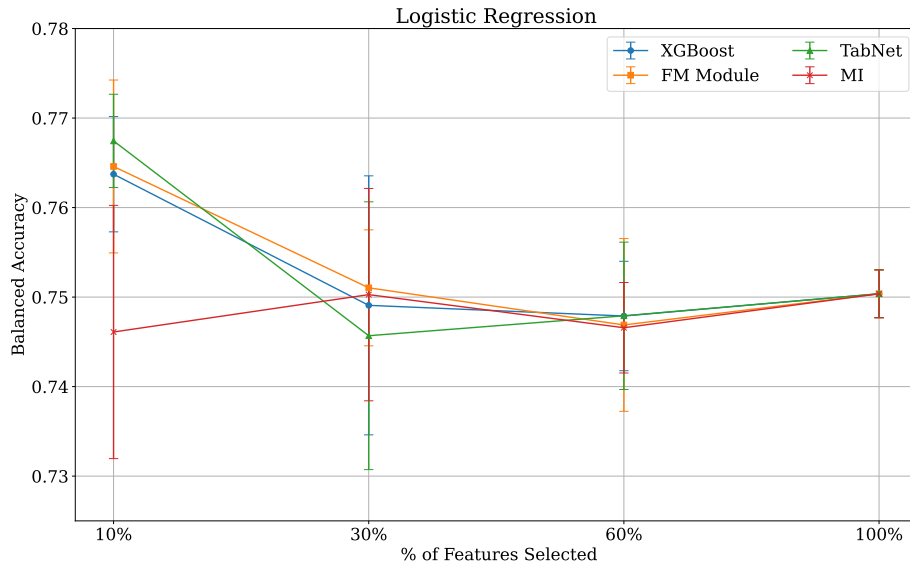


Figure 4.10.: Plot of balanced accuracy obtained for various percentages of top ranked features in logistic regression on NAKO dataset.

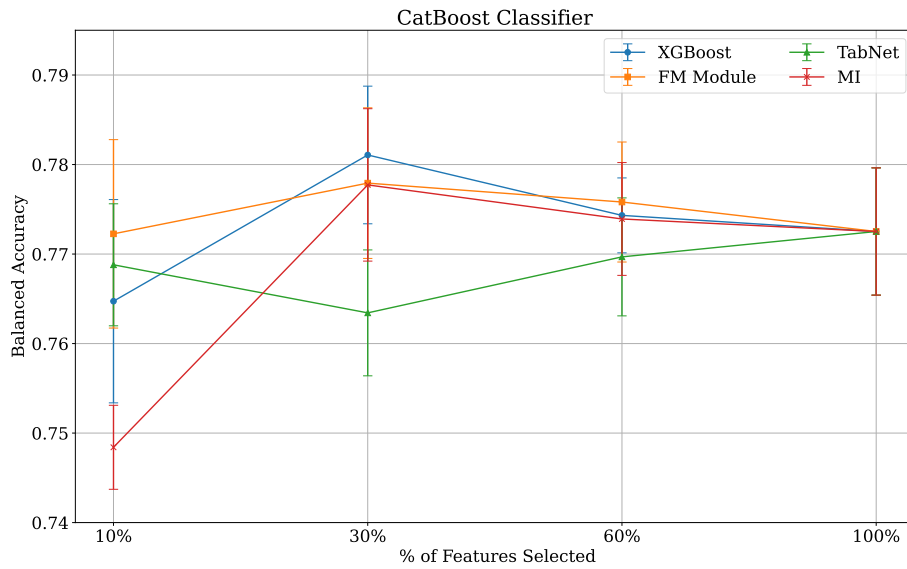


Figure 4.11.: Plot of balanced accuracy obtained for various percentages of top ranked features in CatBoost classifier on NAKO dataset.

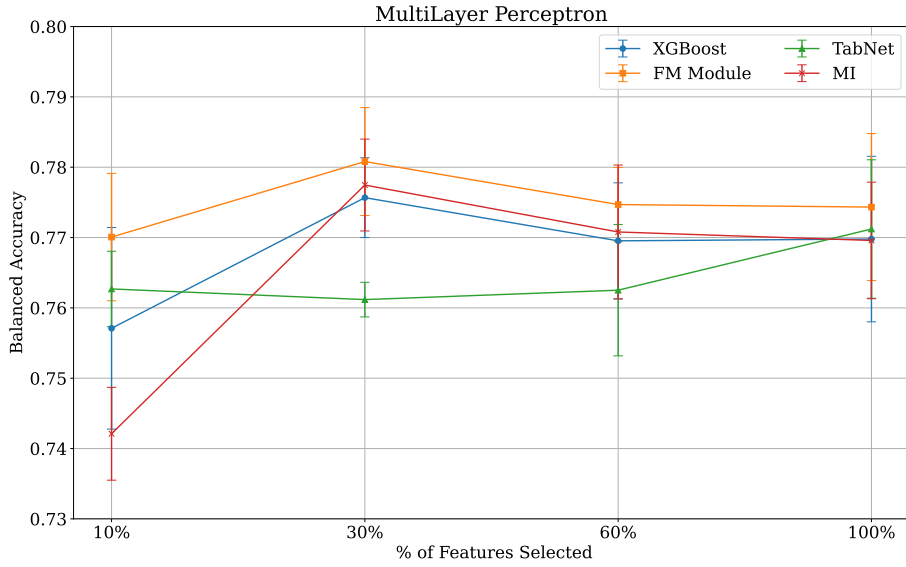


Figure 4.12.: Plot of balanced accuracy obtained for various percentages of top ranked features in MLP classifier on NAKO dataset.

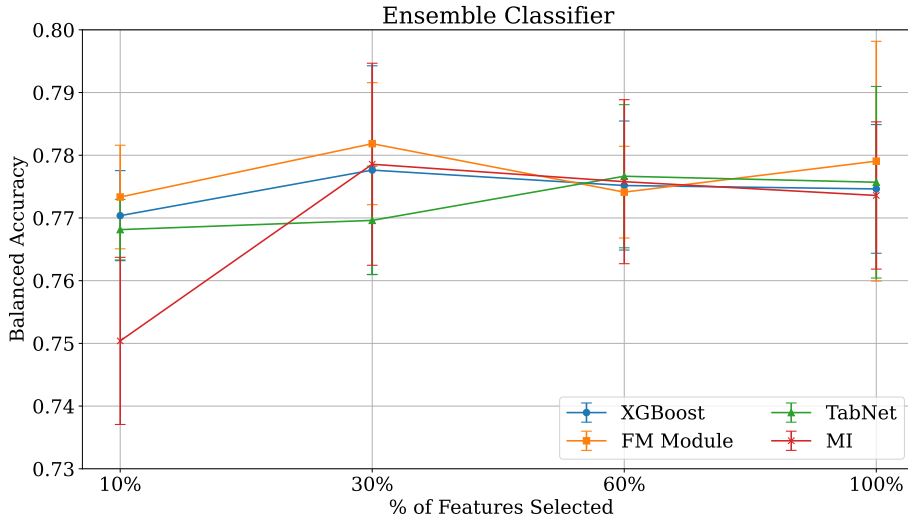


Figure 4.13.: Plot of balanced accuracy obtained for various percentages of top ranked features in ensemble classifier on NAKO dataset.

Both MLP and ensemble classifiers exhibited similar trends, achieving optimal performance on NAKO dataset with a subset of 25 features. These classifiers slightly outperformed CatBoost in terms of accuracy highlighting the capacity of deep neural networks to learn the non-linear feature interactions more effectively. Feature subsets selected by FM Module and mutual information ranked among the top performers for these models.

While mutual information-based selection underperformed at very low feature thresholds

(e.g., 10%), its performance improved substantially when the subset size increased from 10% to 30%. At the 30% threshold, the features selected using mutual information demonstrated comparable predictive performance to those selected by more sophisticated model-based methods across most classifiers. This suggests that although mutual information is a univariate feature ranking method which may struggle to capture interactions in small feature subsets (10% of total features), it can still be effective when provided with a broader subset size ($\geq 30\%$ of total features).

Overall, the FM Module demonstrated consistent performance across all feature subset sizes, making it a reliable method for identifying informative features. The mutual information (MI) method, while underperforming at lower thresholds, appeared to capture most of the relevant features at the 30% mark, suggesting a delayed but ultimately competitive contribution to feature selection.

XGBoost-based rankings showed particularly strong alignment with boosting-based classifiers such as CatBoost, reinforcing the idea that feature selection methods tailored to the structure of a model can enhance downstream performance. In contrast, TabNet a deep learning model specifically designed for tabular data yielded relatively weaker results in feature ranking. This underperformance might be because, TabNet typically benefits from larger datasets to effectively learn meaningful feature attributions[54], which might be lacking with the NAKO datasets. The use of sparse attention and feature masking, can be unstable or overly selective when applied to smaller datasets or limited feature subsets.

The results from both datasets underscore the importance of effective feature ranking methods. The BRFSS dataset comprises a smaller number of well-curated features that are directly related to the target variable, making feature ranking less critical. In contrast, the NAKO dataset includes a large number of features, many of which are weakly informative or noisy. In such cases, identifying complex feature interactions becomes essential for building accurate predictive models. This discrepancy highlights why high classifier performance can still be achieved on the NAKO dataset using a small, well-ranked subset of features. Feature ranking not only improves model interpretability but also enables the identification of potential risk factors for the target condition among numerous clinical or demographic variables. Moreover, selecting a reduced yet informative feature subset can contribute to the computational efficiency and optimization of machine learning algorithms, particularly in high-dimensional medical datasets.

The importance of selecting informative features becomes evident when comparing the performance of feature ranking methods against randomly selected feature subsets of the same size. As shown in Table 4.1, balanced accuracy values obtained using the MLP classifier indicate that random subsets with 8 and 25 features yield notably lower performance compared to subsets selected based on feature importance. This highlights the effectiveness of the proposed feature ranking methods in identifying relevant predictors that contribute meaningfully to the model's performance.

Table 4.1.: Comparison of average balanced accuracy for random features vs importance based feature selection across different feature subset sizes (8, 25, 50, 85 features) on MLP classifier. Best values are in bold.

	8	25	50	85
Random Features	0.635	0.677	0.751	0.771
Feature Ranking Methods				
XGBoost	0.756	0.774	0.770	0.770
FM Module	0.770	0.781	0.776	0.776
TabNet	0.763	0.762	0.764	0.773
Mutual Information	0.744	0.777	0.770	0.770

4.2. Similarity of Feature Ranking Methods

Features that are consistently ranked higher across multiple methods are likely to be more robust and reliable. Therefore we examine the agreement among the top 25 features selected across different ranking methods. The results of the the inter-method agreement is discussed in this section.

RQ.3 How similar are the different feature ranking results?

the similarity of all the features ranked in BRFSS dataset are shown in Figure 4.14 and the similarity among the top 25 ranked features for NAKO dataset is shown in Figure 4.15.

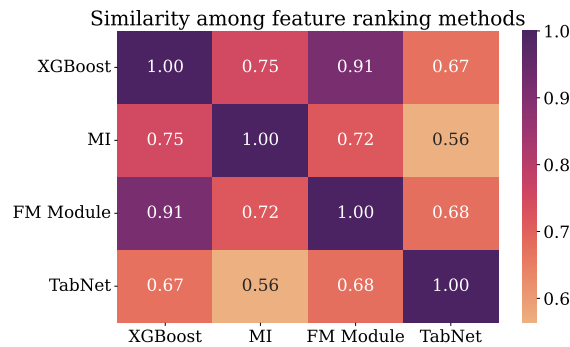


Figure 4.14.: Heatmap showing the pairwise Spearman correlation between feature ranking methods on top 21 features in BRFSS dataset

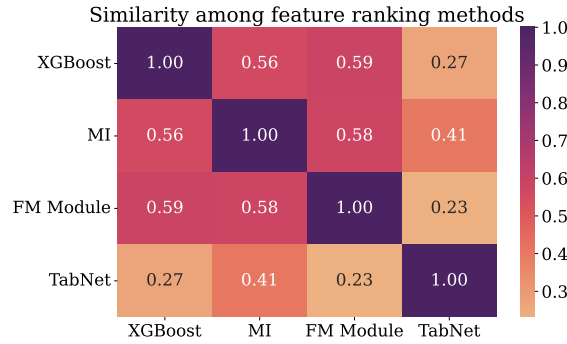


Figure 4.15.: Heatmap showing the pairwise Spearman correlation between feature ranking methods on top 25 features in NAKO dataset

The similarity matrices for the BRFS and NAKO datasets reveal significant different correlation scales. BRFS dataset with a small feature space, all pairwise correlations among feature ranking methods are relatively high (for example, XGBoost vs. FM Module = 0.91). In contrast, even though in NAKO, the analysis was restricted to the top 25 features, the NAKO matrix shows much lower correlations overall (XGBoost vs. FM Module = 0.59). This discrepancy can be attributed to the difference in feature space dimensionality. The large number of candidate features (total 84 features) introduces greater variability in important scores. As a result, different ranking methods tend to agree less perfectly on the feature ordering, explaining the reduced agreement. Naturally, we therefore focus on relative patterns of agreement (which method pairs have higher correlation) rather than on the raw coefficient magnitudes.

FM Module and XGBoost feature rankings exhibit greatest similarity on both datasets, suggesting that these two methods tend to prioritize features quite similarly. A plausible reason is that both methods inherently favor features which has high signal to noise ratio and are consistently informative across all batches (random subsets in case of boosting). This explains why classifiers trained on XGBoost selected versus FM module selected features tend to exhibit similar accuracy trends as shown in 4.11. Since the underlying feature sets contain the same high-impact variables, they exhibit parallel downstream behavior.

A moderate correlation was observed between the MI based feature ranking and other methods. MI method prioritizes features that provide the maximum information about the target variable (information gain). It is obvious that many of these key informative features will also appear in the rankings of XGBoost and FM since they too seek predictive strength, which explains the moderate correlation. However, MI does not incorporate the model-driven biases such as capturing feature interactions or optimizing model loss. As a result, its rankings do not align as closely with those of XGBoost or the FM Module.

The lowest correlation in feature rankings is observed with the TabNet model. Although TabNet shares some algorithmic principles with XGBoost (such as sequential decision steps) and, like the FM Module, it is based on deep learning, it may require a larger dataset to

effectively capture complex feature interactions. The relatively low agreement suggests that TabNet’s feature selection behavior is more sensitive to data size and structure compared to the other methods.

4.3. Selection of Hyper parameters

The selection of hyper parameters are very important to have a well tuned feature selection subsets . This section briefly explains the hyperparameters selected for each of the feature ranking methods and their impact on the ranking algorithm.

XGBoost Feature Importance: To avoid overfitting during feature ranking, shallow trees with a maximum depth of 3 were used. Additionally, the regularization parameter gamma was employed to control the complexity of the model by specifying the minimum loss reduction required to make a further split. Gamma is particularly effective when used with shallow trees, as it helps prevent unnecessary splits and enhances generalization.

Table 4.2.: Hyperparameters for XGBoost feature importance analysis on NAKO dataset.

Parameter	Value
Booster	gbtree
Number of Estimators	50
Maximum Depth	3
Learning Rate	0.2
Subsample	0.7
Gamma	0.2

Feature Importance based on FM Module: The FM module consists of two fully connected layers with 8 neurons in each layer. The choice of 8 neurons strikes a balance between model capacity and regularization. This low-dimensional representation encourages the model to focus on the most informative combinations of input features, supporting its role in feature ranking and selection. Unlike its deeper feature ranking counterparts, one of the highlights of the FM module is its relatively small number of hyperparameters, which simplifies model tuning.

The hyper parameters used in learning network for NAKO dataset is given in Table 4.3. Among these, the batch size was particularly important, as feature importance depends on the primitive feature mask averaged over each mini-batch. Feature selection performance was evaluated over batch sizes of 256, 512, 1028, and 2500. Smaller batch sizes resulted in worse downstream performance on selected features compared to larger batch sizes. This can be attributed to the fact that larger batches produce more representative transformations of the data. Batch size of 1028 was selected for the NAKO dataset, while 2500 was used for the BRFSS dataset.

Table 4.3.: Hyperparameters for learning network $g(\cdot)$ of FM module based feature ranking on NAKO Dataset.

Parameter	Value
Number of layers	4
Dropout	0.3
Learning Rate	1e-3
Epochs	40
Batch Size	1028

TabNet Feature Importance:

Table 4.4.: Hyperparameters for TabNet based Feature Importance on NAKO Dataset

Parameter	Value
Number of decision steps	3
N_d & N_a	8
Mask Type	sparsemax
Batch size	1024
Virtual batch size	128
Relaxation Factor	1.5

According to [33], for datasets with approximately 10,000 samples, the parameters N_d and N_a were set to 8. While this configuration yielded good performance, slight overfitting was observed during training. Reducing the values of N_d and N_a to 4 led to a noticeable drop in accuracy. Similar to the FM-based feature ranking method, both the batch size and the virtual batch size had a significant impact on the model's performance. In particular, it was observed that lower virtual batch sizes resulted in training instability.

5. Conclusion & Future Work

This work aimed to identify potential risk factors or biomarkers in disease prediction by leveraging the predictive power of machine learning models, going beyond conventional statistical methods commonly used by medical practitioners. Traditional statistical approaches often have difficulty capturing complex feature interactions and usually depend on domain expertise both to select relevant features and to validate the results.

One of the key findings of this study is the identification of novel risk factors such as fat fractions in thigh muscle groups and indicators of physical fitness of individuals which may indicate the early development of diabetes. This is in addition to the well known factors such as BMI, age and other metabolic syndromes. Comparison across different methods revealed that model-based feature ranking approaches might be slightly biased to the specific learning algorithm used. To address this, multiple feature selection techniques were employed and features consistently that ranked across methods were discussed.

Among the approaches, deep learning based feature mask method derived from [32] demonstrated consistent superiority in finding the informative features closely followed by other methods. This was evident from the performance of downstream classifiers trained on selected feature subsets, highlighting the effectiveness of leveraging deep learning techniques for feature rankings in tabular datasets. Complex models like TabNet showed a slight overfitting in our experiments, they may prove effective in larger datasets and warrant further investigation.

This work examined two distinct datasets with a common target objective, supported by extensive visualizations and cross-validation of the results to ensure reliability. The identification of key risk factors for disease through machine learning can assist medical practitioners in prioritizing relevant features during regular screenings. By enabling earlier detection and more targeted assessments, this approach not only enhances diagnostic efficiency but also supports timely, proactive intervention ultimately improving patient outcomes.

5.1. Future Work

This work can be further extended by ensembling multiple feature ranking methods which may improve the reliability and robustness of the results by reducing model-specific biases. Additionally, incorporating other advanced methods such as Minimum Redundancy Maximum Relevance (mRMR) instead of mutual information could further enhance the quality of feature selection by considering both relevance and redundancy. Additionally, SHAP (SHapley Additive exPlanations) analysis could be employed as a post-hoc interpretability method to provide deeper insights into the model's decision-making process and validate the importance of the selected features.

The current analysis was based on static, tabular data. Given that the datasets originate from patient-based surveys, collecting data at regular time intervals on the same individuals could enable longitudinal analysis. This would provide insights not only into the identification of risk factors but also into their progression and the development of disease over time.

Future extensions could include multimodal learning by integrating image data (e.g., MRI scans) with tabular data. Such combined analyses have the potential to improve model performance while offering a more comprehensive understanding of the patient's condition. Although this integration may complicate the learning process and pose challenges to interpretability, exploring interpretable multimodal learning frameworks in this context presents a promising direction for future research.

A. Additionally

A.1. Description of Features used in the Experiments

Feature Name	Description	Type
anthro_groe	Height	Numeric
anthro_gew	Weight	Numeric
d_an_met_1	Diabetes mellitus	Binary
sa_hba1c	HbA1c: measured value, in mmol/mol	Numeric
sa_ogtt0	oGTT Glucose 0h: measured value, in mmol/l	Numeric
sa_ogtt2	oGTT Glucose 2h: measured value, in mmol/l	Numeric
ogtt_o3	Are you currently being treated with insulin or blood sugar lowering tablets?	Binary
BMI	Body Mass Index	Numeric
d_an_cv_6	High blood pressure	Binary
d_an_met_2	Elevated blood lipids, cholesterol, or triglycerides?	Binary
basis_age	Age	Numeric
anthro_reaktanz	Reactance Xc 50 kHz	Numeric
anthro_resistanz	Resistance R 50 kHz	Numeric
anthro_fettmasse	Fat mass in %	Numeric
anthro_taillenumfang	Waist circumference	Numeric
FF_Glu	Fat fraction in gluteus muscle group (average of left and right thighs)	Numeric
FF_Add	Fat fraction in adductors muscle group (average of left and right thighs)	Numeric
FF_Ext	Fat fraction in extensors (average of left and right thighs)	Numeric
FF_Pso	Fat fraction in psoas (average of left and right thighs)	Numeric
CSA_Glu	Maximum axial cross-sectional area of gluteus muscles	Numeric
avg_vat_ultra	Average abdominal fat distribution (ultrasound measurements)	Numeric
spiro_mw_fvc	Forced expiratory vital capacity (FVC)	Numeric
spiro_mw_fev1_fvc_p	FEV1 / FVC - target value	Numeric
ws_rk_extension	Range of motion: extension (right knee joint)	Numeric
ws_lk_extension	Range of motion: extension (left knee joint)	Numeric
ws_rh_innenrotation	Range of motion: internal rotation (left hip joint)	Numeric
d_qol3_1	Moderately difficult activities	Ordinal
d_fit	Physical fitness	Ordinal

A.2. Visualizations

Visualization of diabetes distribution based on the top ranked features in BRFSS dataset

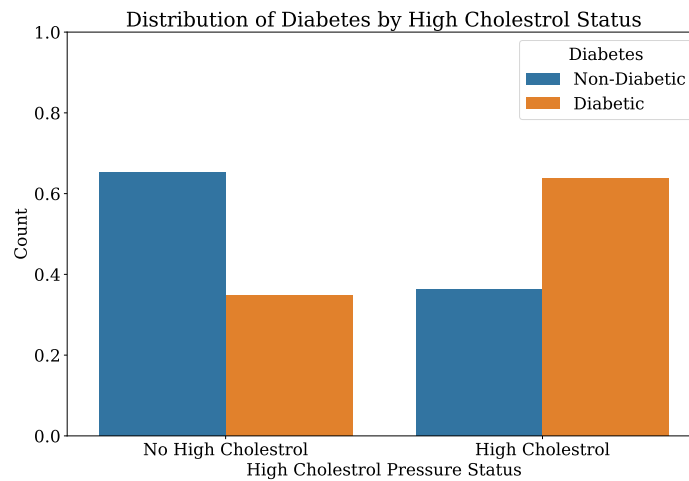


Figure A.1.: Bar plot showing the distribution diabetic individuals based on cholesterol status. Individuals with high cholesterol were more prone to Diabetes.

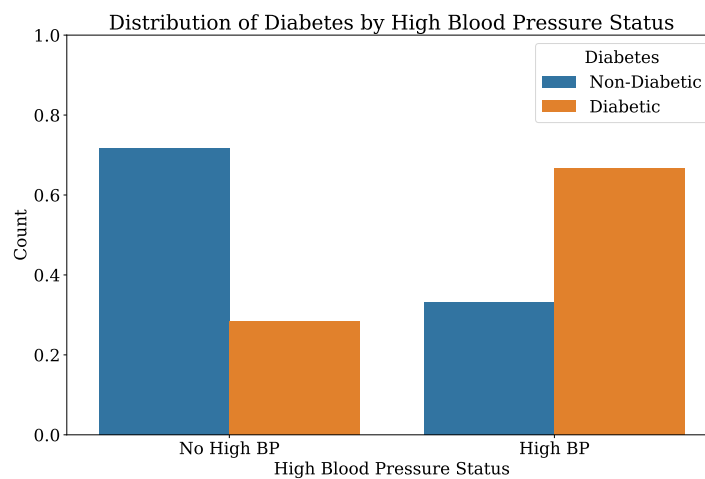


Figure A.2.: Bar plot showing the distribution diabetic individuals based on Blood Pressure status. Individuals with high blood pressure were more prone to Diabetes.

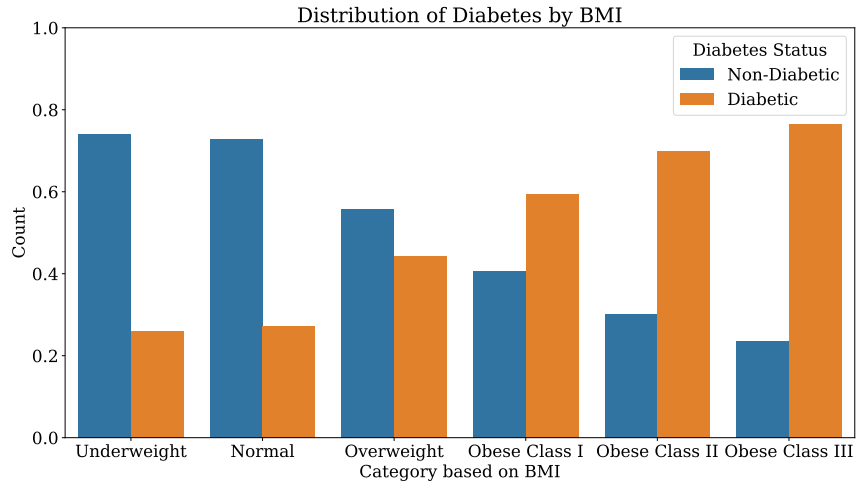


Figure A.3.: Bar plot showing the distribution diabetic individuals based on BMI Categories. Individuals with High BMI (obese category) were more prone to Diabetes.

Visualization of diabetes distribution based on the top ranked features in NAKO dataset

The difference in fat fraction of gluteus muscles is larger compared to other muscle groups. Therefore, it always ranked above all other measurements derived from MRI results. This is consistent among all feature ranking methods.

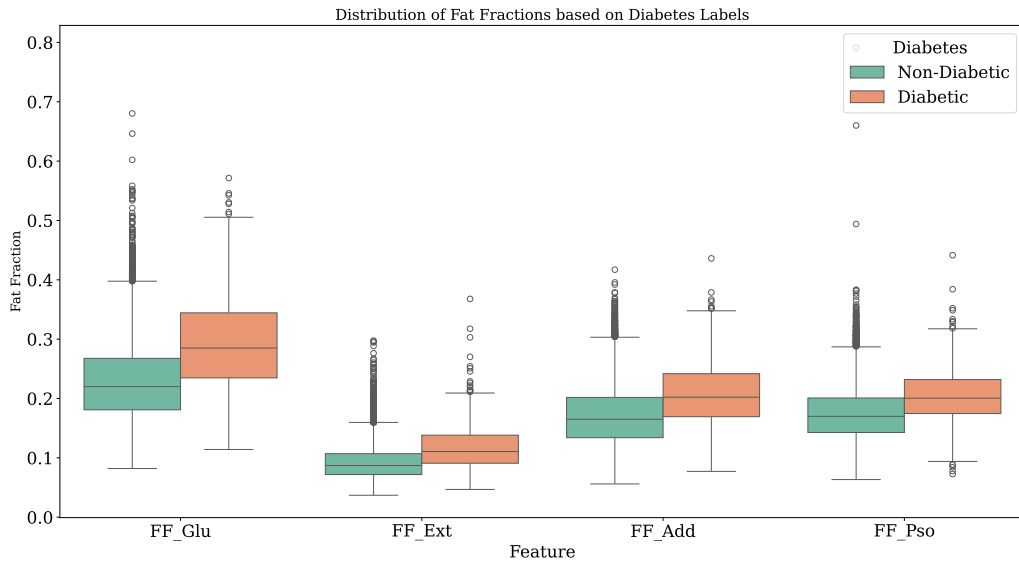


Figure A.4.: Box plot showing the distribution of fat fractions in thigh muscles (Gluteus, Psoas, Adductors and Extensors) based on Diabetes labels. Mean fat fraction is higher in diabetic individuals.

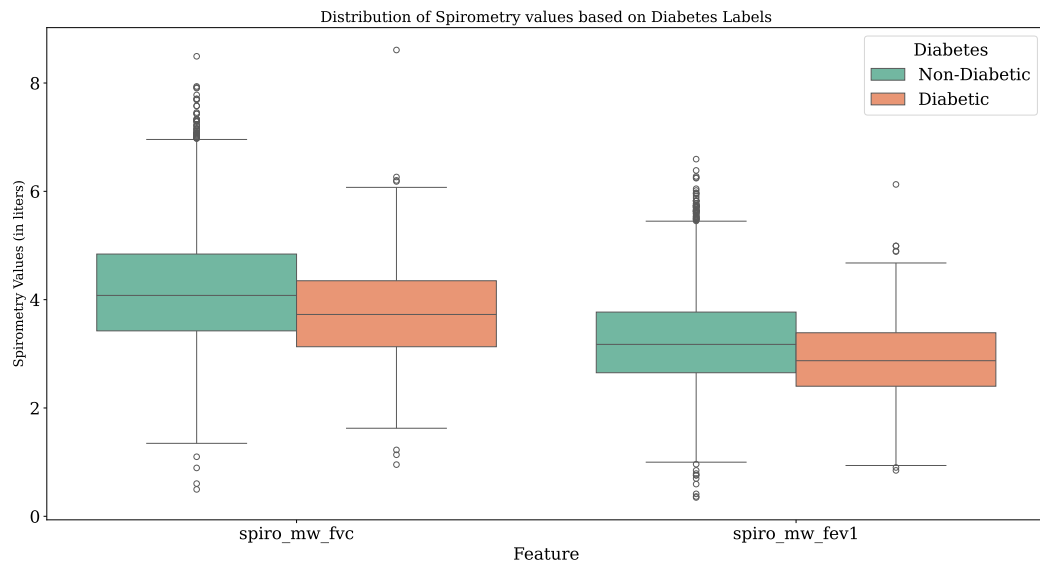


Figure A.5.: Box plot showing the distribution of spirometry values based on Diabetes labels. Functional capacity is generally lower in diabetic individuals

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Declaration

Herewith, I declare that I have developed and written the enclosed thesis entirely by myself and that I have not used sources or means except those declared.

This thesis has not been submitted to any other authority to achieve an academic grading and has not been published elsewhere.

Stuttgart, 01.05.2025



Veena Sridhar